

Quail Springs Permaculture, Spring 2022 PDC

Permaculture Design Project Workbook



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Design Project: You will work on your individual permaculture design project throughout the course by completing assignments in this workbook, ideally each week. The purpose of the individual design project is to act as a demonstration of your learning through creating a site design that uses the guiding permaculture ethics and integrates permaculture principles throughout.

Printing the Workbook: You are welcome to print this workbook if you prefer to work by hand. If you choose to print your workbook, you will need to upload your notes and drawings into this workbook regularly so we can see your work.

Visual vs. Written Work: As you read in the *Design Project and Process* lesson in the course curriculum, the majority of the visual mapping components of this workbook are

optional (except the last section which will be required for a certificate). This is unusual, as traditionally PDC design projects are mostly visual.

If choosing to use no visual imagery, please try to use as much written imagery and detail as possible in their assignment narratives. In other words, try to “show your work” through your writing in as much detail as possible.

Note: If you want your design reviewed & commented on and in order to receive a Permaculture design certificate, a visual design is required (in the last section 12, Part 4).

Maps: When you draw maps, you can draw them by hand and upload photos/scans into this document, or use digital tools such as an iPad or a collaborative whiteboard such as Miro, and then import that to your workbook.

Location: After you email Aris the link to your workbook, it will be stored in a shared QS PDC drive so that QS staff can access it. At the end of the course, you are welcome to copy your workbook and use it for other personal projects.

Weekly Reflections: Part 1 of the design workbook is mostly a collection of *observation* and *assessment* assignments of your site with the goal of guiding you towards a deep understanding of your site. This is so that once you move into the conceptual design of your site in Part 2 you will have enough intimacy with your site to create a well-integrated design in Part 3.

The **weekly reflections** are a space to *vision* about what your future design plans for your site will look like. We highly encourage you to use your weekly reflection space to vision and dream about possibilities for your final design without consequence. Your weekly reflections are a wonderful place for you to track your process as you move through this learning journey.

Credit: This workbook is built upon the knowledge of generations of land tenders and teachers. We have protected the workbook under copyleft licensing that allows you to use and modify this work, while requiring that copyleft be maintained for derivative works created from it. This workbook is inspired by work originally formatted by permaculture teacher Andrew Millison of OSU, then re-adapted by permaculture teacher Daniel Parra-Hensel of SBCC. We have altered and adapted this work for the Quail Springs course.

Help: If you have questions about your project, please post them in the #2022pdc-designproject Slack channel.

Before we start - it is helpful to think of this workbook (and permaculture design in general!) as an exploration of constraints and opportunities. As designers we want to

seek to understand, out of a set of things, which one will influence the rest. And then dive deeply into designing around the thing that constrains the rest, because that element is your biggest driver. For example, if a group of people decide to buy a boat together, there are a number of things they need to consider. They will have to decide how big they want the boat to be, where to keep the boat, and what days certain people get to use it. But, if they have a limited budget, that will influence all of those things - so the group will have to begin at that constraint and design for that factor as the most influential part of their plan. They will then uncover all the opportunities available as they work from that constraint. Keep this framework of constraints and opportunities in mind as you dive into the assessment, visioning, and design of your site.

Part 1

PDC Weeks 2-12

1. Choosing A Design Site for the Permaculture Design Course

Strategies for Success

In this course you will be designing your very own project site! You will be creating an integrated design plan for a piece of land that you have access to. The site could be the home where you live, a community-based project, a school garden, a park, or your workplace. This should be a place that you are committed to developing a relationship to over the duration of the course. It should be convenient for you to access, ideally within walking distance from where you live. The following six guidelines will support you in choosing a design site that will guarantee success and to gain the most from this course.

Please review the following guidelines:

- Be sure that you are **able to visit the design site frequently** during the course. A big part of the design process is site analysis which is based on the permaculture principle P.A.T.O. (protracted and thoughtful observation), meaning both past and current observations. We will be asking you to spend time observing a variety of aspects related to the site throughout the 24 weeks of this course.

- Be sure that **you can dig holes in the soil on the site**. One assignment is a soil analysis which includes digging holes in the soil to take samples and test drainage.
- Since some assignments include analyzing how structures interact with the land on the site, we ask that you choose **a site with at least one building**. If that's not possible you can plan a structure into your final site design.
- Be sure you have the **ability to communicate with the owner, or those with decision making power** of the property. Practicing the process of interviewing the client/community and using what you have learned to develop your design is essential if you are aiming to do design work in the future.
- It's best if the **possibility of implementation on the site is real**. This allows you to practice working within an actual budget and designing for success with the resources at hand. If this is not possible, a theoretical design can work, but we encourage you to choose a site that you have the potential to carry through from design to implementation.
- We ask that you **use the entire area within the property boundary** as your site for the course work. While you may be interested in focusing on one particular area, in this course we encourage you to look at the entire area when doing site analysis. This will allow you to consider how to integrate the challenges and resources from the entire property into a design that meets the project goals. You may choose to focus on one or more smaller areas and create map insets that allow you a higher level of detail in your design work.

Now that you have chosen your site let's get into the ***design site survey***.

2. Design Site Survey

What is the address of your site? **10239 Ojai Santa Paula Road**

What are the GPS coordinates? **34.437853, -119.167298**

What is the nearest town or city center? How far is it?

Ojai, CA 6 miles away

What is the size of your site in acres, hectares, square feet or square meters?

5.5 acres

Look at your site from different mapping platforms. Try [Google Earth](#), [Google Maps](#), [Bing Maps](#), etc. Try toggling the view to see different perspectives on your site.



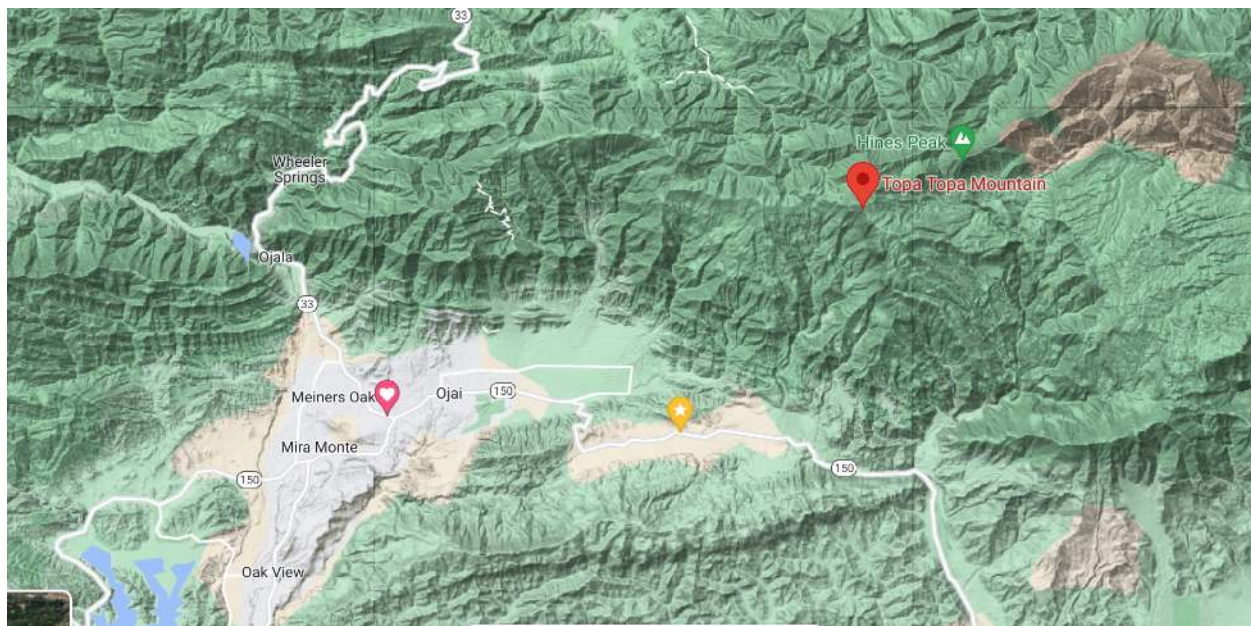
North is UP



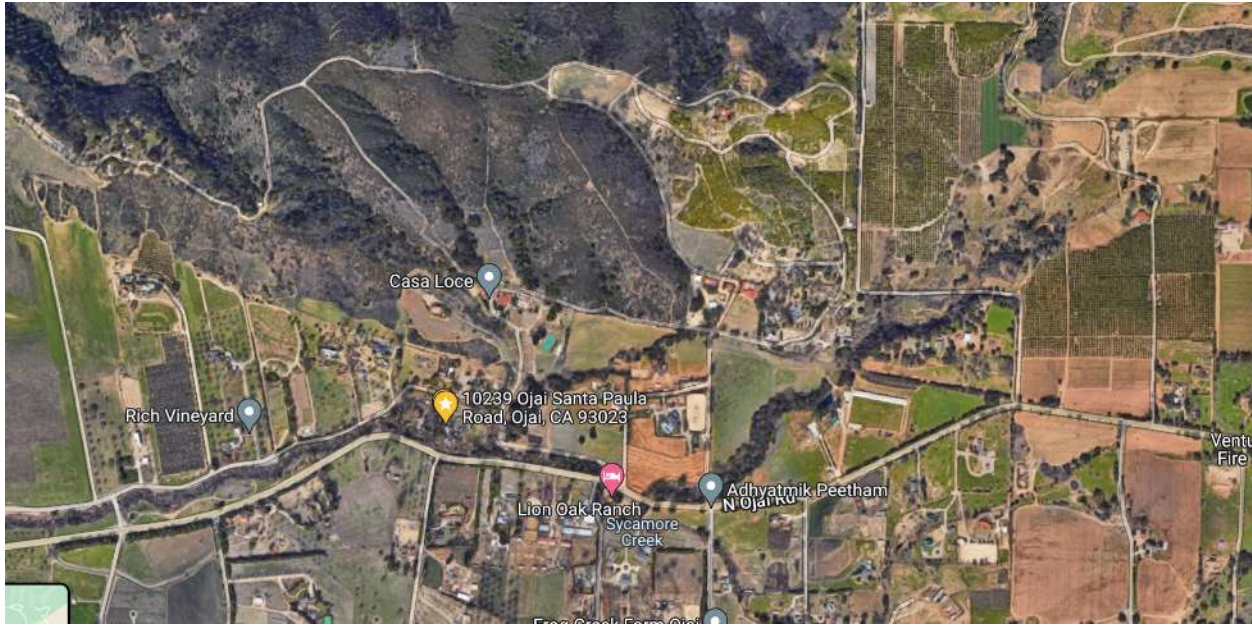
Perspective from the WEST



Perspective from the NORTH



Topo Map



Wide View



Contour Map

The blue area at 1308 feet should continue across the property from left to right. That is where the Lion Creek is. There is a

discrepancy between this map and Google Earth map which gives a range of 1240 to 1300 feet.

If you're interested in deeper research you can dive into property maps (for US students) from your specific county online which often can depict more accurate property lines and other information.

Share what you learned about your site from this exercise:

Full circle Farm is located in Upper Ojai at about 1,250 feet altitude (google maps figures). to the north is the Padres National Forest, which Rises to 6400 feet. There are lots of farms in the area and Orchards too.

Please import photos or illustrations of your site, ideally looking towards the highest point, lowest point, or any major features of importance (incl. descriptive orientation so

photos/illustrations can be cross-referenced to each other and a basemap/aerial view).



Main House

This is the center of the community, where the cooking and meetings take place. Four people live in the house too. Zone 1



Garden Beds

The garden is surrounded by a fence to keep critters out. Zone 2



View South from the Hill

This is the view from the top of the water tank. You can see persimmon trees in the foreground and the garden in the background. Zone 5 is in that direction too. The elevation gain is about 25 feet.



Chicken Koop

Here's where the chickens are, looking South, and next to the garden. The trash area is to the right. Zone 2



Chickens

These layers produce about 30 eggs daily.



Water Tank

This is a 10,000 gallon water tank. Actually we have 2 of these at the top of the hill on the North side. I'm not totally sure how to use them as we already have well water. Filling them requires a lot of water that is easily wasted when someone leaves a spigot on, which happens all the time. Zone 2



Well House

We have a 200 foot well that is solar powered. It supplies all our water needs. We do have city water for backup that we sometimes use.



Persimmon Hill

This view looks up persimmon hill. There are black locust trees and eucalyptus trees too. Zone 2



Zone 4

This is the north east side of the property. It borders the neighbors.



Zone 5

This land is to the south and is where the Lion creek is. It drops down about 20 feet. It's pretty inaccessible.



Zone 5 Southwest

South East corner of property that slopes down to the creek. This is a large area where no one goes. There have been bobcats down there and probably lots of racoons and possums



Zone 5 Southeast

This is Zone 5 land that goes down to the creek. Nobody goes down there as it's too dense with brush and the ground slopes



West Orchard

This is the front orchard with orange, lemon and persimmon trees.

Zone 2



Garden Shed

A shady spot to put gardening tools and pot plants. To the left is a greenhouse, but it gets way too hot in the summer.



Tiny Home

Converted from a cargo trailer.



Cabin

This is a studio cabin with a nice deck and a single room on the inside.



Earthdome

Architecture so strong it withstood a forest fire.



Large Cabin

This cabin rents for \$900 a month. It has a basic kitchen and 3 small rooms on the inside.



Barn

Used for storage. The roof needs work. I had to reinforce it after a storm.



Stage and Community Room

Our newest construction project used to be a dirt floor. We have lots of social events so this gets a lot of use. It's next to the Workshop.

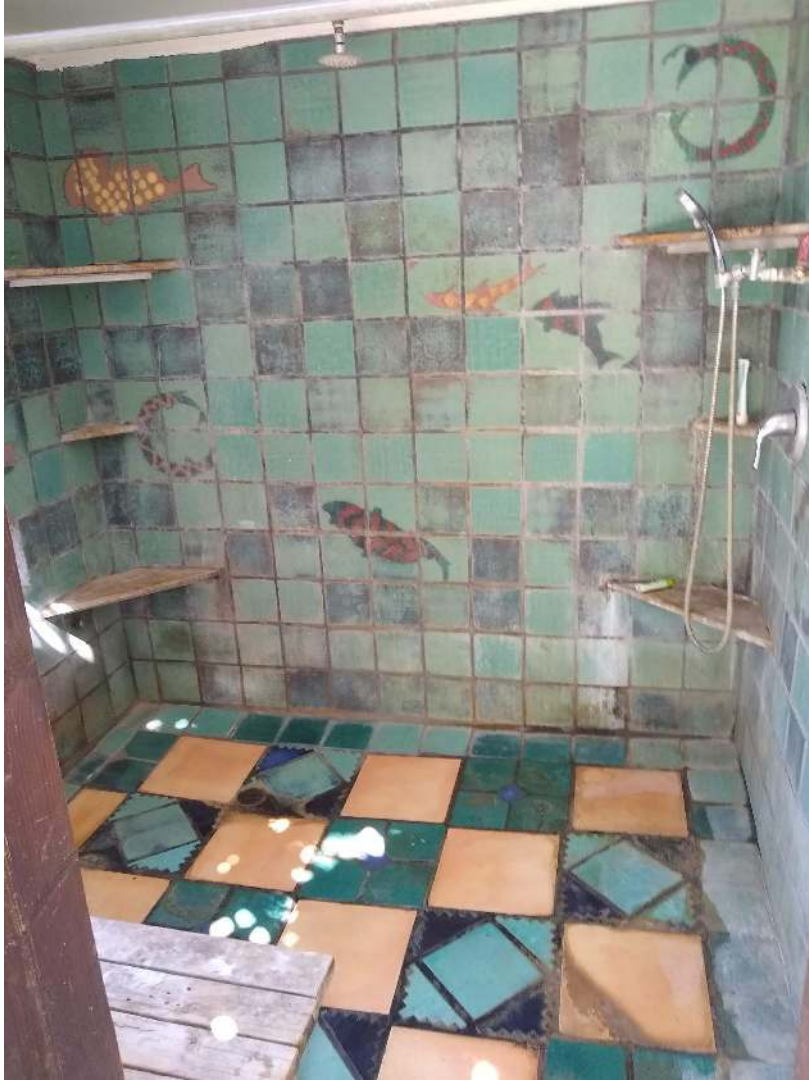


Workshop

It gets a lot of traffic and use



Laundry and Bathhouse
More options for gray water.



Shower in Bathhouse

This feeds directly into the arundo patch, but I'm going to create a greywater system with branch drains.



Gym

Built with mostly leftover material and donated equipment. It gets a lot of use.



Tractor

Our newest acquisition. With backhoe and chipper.



Farm Truck

Used for mulch, lumber and supply runs to town.



Pomegranate Trees

We have about 15 of these trees. We had more but a lot were destroyed in the Thomas Fire in 2017. They fruit in the winter time.



Bee Hive

Not only for honey but bee sting therapy.



Worm Bin

Full of red wigglers. We make compost tea by the bucketful!



Compost Pile

It's just an unmaintained pile

2.1. Mapping through Visual or Writing

Written option:

Describe your site in detail including where all the above features are located and any additional details that might impact your site design. Please note how all of these features are oriented (what direction are they facing and how do they relate to one another). Feel free to add pictures as well.

Full Circle Farm has been around for 38 years, founded by Bob Goddard. Bob bought the land for a good price after a fire came through and destroyed the house. He then invited people to live on the land if they would help rebuild the house. And so the community was born. About 20 people live here now along with 3 children on 5.5 Acres. There are cabins, trailers, motorhomes, a main house, a tiny home, an earthbag shelter, a FEMA tent, a geodesic dome, a workshop, barn, and garden shed, several tool sheds and a shed to house the solar powered well pump. In the center at the property towards the east there is the garden which is about 1 acre in total. On the west side there is an orchard with persimmon, orange, lemon and pomegranate trees. On the North side we have Persimmon Hill with 20 persimmon trees. It's the high point of the land, rising up about 20 ft above everything else. Up there we have two large galvanized steel water storage tanks which right now don't get much use.

Mapping option:

Draw a map of your site using different colors and lines (solid vs. dotted) to identify the different features of your site. It does not have to be perfect but please do your best to make it as clear as possible and to scale. Please include a key. Include the orientation of your map using a compass rose to demonstrate north. In the permaculture design process, the process of creating a map that includes all the fixed features of the site is called base mapping. This base map is the starting point of your site, before you add any additionally designed systems on the site. One way designers will often do this (by hand) is to trace a map of their site, and then add the fixed features on top. Other ways include using [Google Earth](#) or other software to do more advanced layering throughout the design process.

Base Map



2.2. Climate Survey

Please answer the survey questions to the best of your ability. You may include photos and images to support your answers where applicable. Please cite sources of data.

First and last average frost date:

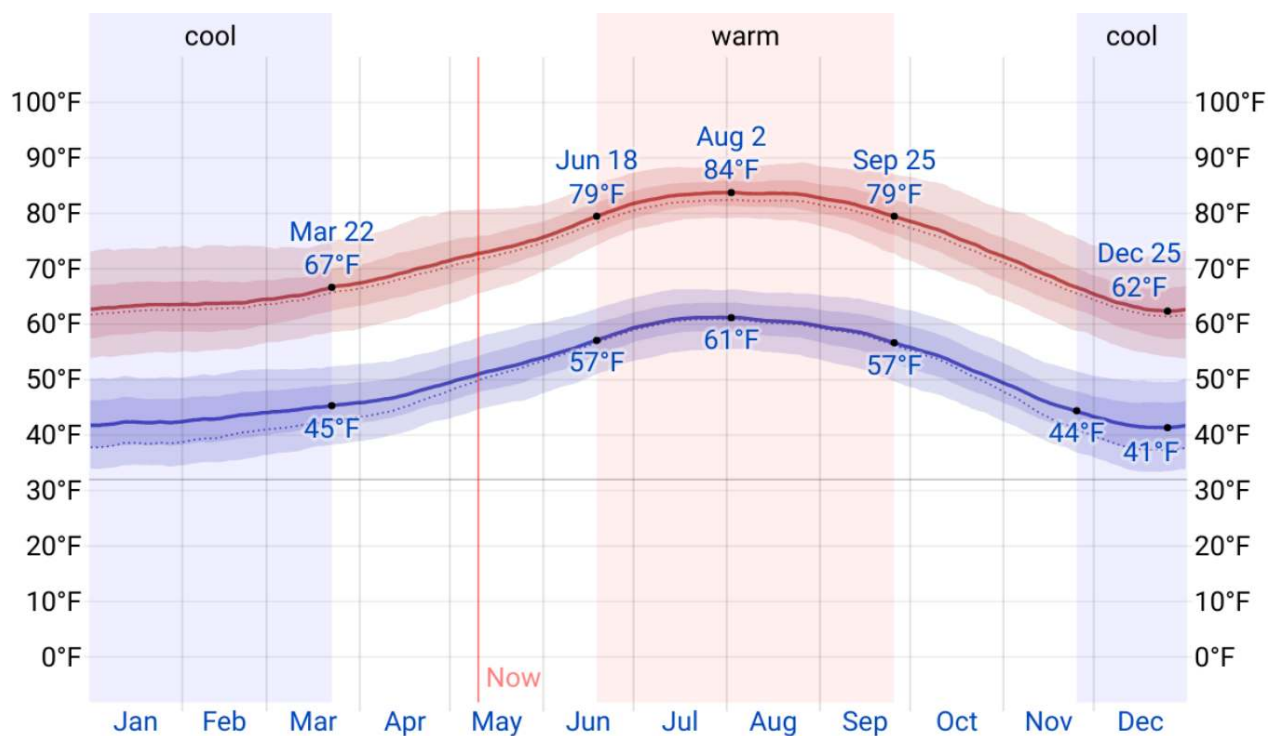
Nearest Climate Station	Altitude	Last Spring Frost	First Fall Frost	Growing Season
OJAI, CA	744'	Apr 3	Nov 7	217 days

A **frost date** is the average date of the last light freeze in spring or the first light freeze in fall. The classification of freeze temperatures is based on their effect on plants:

- **Light freeze:** 29° to 32°F (-1.7° to 0°C)—tender plants are killed.
- **Moderate freeze:** 25° to 28°F (-3.9° to -2.2°C)—widely destructive to most vegetation.
- **Severe freeze:** 24°F (-4.4°C) and colder—heavy damage to most garden plants.

From: <https://www.almanac.com/gardening/frostdates/zipcode/93023>

Number of days of frost-free growing season:



The daily average high (red line) and low (blue line) temperature, with 25th to 75th and 10th to 90th percentile bands. The thin dotted lines are the corresponding average perceived temperatures.

Average	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
High	63°F	64°F	66°F	70°F	74°F	79°F	83°F	83°F	81°F	75°F	69°F	63°F
Temp.	51°F	52°F	54°F	58°F	62°F	67°F	71°F	71°F	68°F	63°F	56°F	50°F
Low	42°F	43°F	45°F	48°F	52°F	57°F	61°F	61°F	58°F	53°F	46°F	42°F

[USDA hardiness zones](#) (US students), non-US students equivalent:

*More detailed [hardiness zone map](#) for California students

9b (25 to 30 °F/-3.9 to -1.1 °C)

Average annual rainfall and average annual snowfall:

Climate in Ojai, California

Ojai, California gets 19 inches of rain, on average, per year. The US average is 38 inches of rain per year.

Ojai averages 0 inches of snow per year. The US average is 28 inches of snow per year.

On average, there are 269 sunny days per year in Ojai. The US average is 205 sunny days.

Ojai gets some kind of precipitation, on average, 35 days per year. Precipitation is rain, snow, sleet, or hail that falls to the ground. In order for precipitation to be counted you have to get at least .01 inches on the ground to measure.

Weather Highlights

Summer High: the July high is around 86 degrees

Winter Low: the January low is 40

Rain: averages 19 inches of rain a year

Snow: averages 0 inches of snow a year

BESTPLACES COMFORT INDEX

The annual BestPlaces Comfort Index for Ojai is 8.9 (10=best), which means it is more comfortable than most places in California.

9.3 / 8.4

Summer / Winter

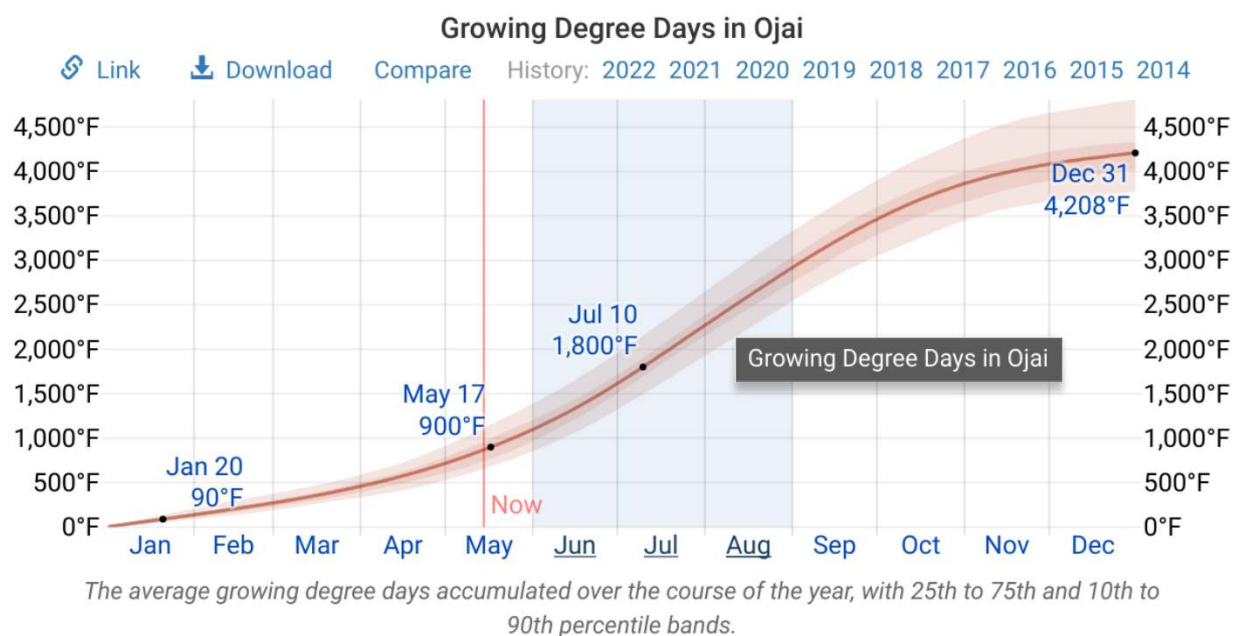
June, July and May are the most pleasant months in Ojai, while January and February are the least comfortable months.

JanFebMarAprMayJunJulAugSepOctNovDec0510BestPlaces.Net

Climate Averages

	Ojai, California	United States
Rainfall	18.9 in.	38.1 in.

Snowfall	0.0 in.	27.8 in.
Precipitation	34.8 days	106.2 days
Sunny	269 days	205 days
Avg. July High	85.5°	85.8°
Avg. Jan. Low	40.1°	21.7°
Comfort Index (higher=better)	8.9	7
UV Index	6	4.3
Elevation	745 ft.	2443 ft.



What is the lowest temperature reached in an average year (in what month?):

CLIMATE

Hot **High Temperature Average** high temperature in °F

August is the hottest month for Ojai with an average high temperature of 86.8°, which ranks it as cooler than most places in California. In Ojai, there are 7 comfortable months with high temperatures in the range of 70-85°. The most pleasant months of the year for Ojai are October, May and June.

Very Hot Days # days where the high temperature rises above 90° F

In Ojai, there are 36.3 days annually when the high temperature is over 90°, which is cooler than most places in California.

Low Temperature Average low temperature in °F

December has the coldest nighttime temperatures for Ojai with an average of 39.3°. This is warmer than most places in California.

Freezing Days # days where the low temperature falls below 32°

In Ojai, there are 13.1 days annually when the nighttime low temperature falls below freezing, which is about average compared to other places in California.

Extremely Cold Days # of days where the low temperature falls below 0°

In Ojai, there are 0.0 days annually when the nighttime low temperature falls below zero°, which is about average compared to other places in California.

Humidity Percentage of the amount of water vapor in the air

Humidity is low in Ojai, and some people might find the dryness to be uncomfortable. Days with noticeable humidity are rare throughout the year. A very dry climate can lead to cracked, itchy skin and dry eyes.

Rainfall Average rainfall in inches.

February is the wettest month in Ojai with 4.8 inches of rain, and the driest month is July with 0.0 inches. The wettest season is Spring with 63% of yearly precipitation and 1% occurs in Autumn, which is the driest season. The annual rainfall of 18.9 inches in Ojai means that it is about average compared to other places in California.

Rainy Days number of days with over 1/10 inch of rain.

February is the rainiest month in Ojai with 6.4 days of rain, and July is the driest month with only 0.3 rainy days. There are 34.8 rainy days annually in Ojai, which is less rainy than most places in California. The rainiest season is Spring when it rains 50% of the time and the driest is Autumn with only a 3% chance of a rainy day.

All time record low temperature (what year?):

29 degrees in 2017

What is the highest temperature reached in an average year(in what month?):

113 Degrees in July

All time record high temperature (what year?):

118 Degrees in 2020

3. Client Interview

The following questions are designed for the person whose site you are designing for. In some cases this may be you. We know that your design site may not have the possibility for implementation (i.e. it might be totally conceptual) and that's okay. Have fun with this.

As part of the Client Questionnaire and Client Desired Outcomes, please include inspiring design images (aka mood board). These could be collected from the internet, magazines and photos. This will help you explore options for images of desired outcomes and learn how to use visuals to communicate and share your ideas. It also helps clients and teachers understand your ideas by adding visual descriptions to your verbal descriptions.

Client Profile

Client names and ages:

Full Circle Farm Community of 20+ adults

Property owners names and ages:

Bob Goddard

Does the client rent or own this site?

Bob is the Owner, Mortgage free

How long has the client lived/owned/tended this site (land)?

37 years

Your relationship to the clients and to the owners (if different)?

The community pays Bob rent for the land, but Bob is an active member in the community too. I am one of the land managers here.

Client occupations:

None. He gets income from investments.

Amount of time available for maintaining property:

We have about 10 people working part time on the land.

Initial budget for implementation:

\$1000/ month

What's the client's dream for the property?

To be a productive community with a garden that feeds all community members.

How long do residents plan to live at this site?
indefinitely.

General lifestyle (active, sedentary, lots of travel etc.):

Half the people who have jobs that bring money into the community. The other half work on the land about 20 hours a week. This consists of working in the garden repairing things, fire clearance, and also cooking and cleaning

Special requirements:

Hobbies: ?

Client's Desired Outcomes

A really nice garden that produces all year round.

Water systems: (What is the existing supply, and future needs for domestic, agricultural, storage?)

1) Existing: Well that goes 250 feet down. City water through a 2 inch pipe, that has very high pressure. Two 10,000 water tanks that don't really get used

2) Desired: better gray water system, and rain harvesting system

Lighting: paths, buildings (existing and future needs)

1) Existing: things are generally well lit.

2) Desired: we are going full solar in the next few months with a roof of solar panels, so we want more path lighting.

Structures: (What are existing and desired structures; i.e. house, greenhouse, shop, barns, chicken house etc.)

1) Existing: We have a main house, a barn, and a workshop. We also have a tiny home, a geodesic dome earthbag Dome FEMA tent, two guest tents, a garden shack, a pump house, a tool shack, free motorhomes, a trailer and 5 cabins. Most of these things are not according to any code.

2) Desired: We probably don't need any more buildings. but we just bought a new tractor so we need a garage for the tractor. And maybe some storage space.

Heating, cooling, electricity: (What are existing sources and desired sources?)

1) Existing: Most of the farm is heated by wood stove oh, because we have a lot of wood

from trees on the land oh, and we can get a lot of wood from anywhere. some people use electric heaters in their places. Nobody uses air conditioning. Electricity is from the grid oh, and it's very expensive. I'm not getting more expensive. We will remove full solar panels later this year.

2) Desired: We could get use more rocket stoves, or rocket mass heaters in some of the community spaces. we're getting solar panels which are good. Also we could use more shade everywhere because summers are really hot and brutal.

Utility areas: clothes line, recycle and trash, wood, service equipment (existing and future needs)

1) Existing: We have a laundry area, oh, that we need to turn into a gray water system. We use dryers and we use clothes lines

2) Desired: I would love to make a pedal human-powered washing machine. I would also like to get people to not use the dryers this much.

Children's areas: (existing and future needs)

1) Existing:

We have a play area by Shines geodesic down. There is an archery place. There used to be an old swing set but it got a little dangerous. There are lots of toys just laying around the property like toy tractors and dump trucks.

2) Desired: We want to expand the children's area with a much bigger playground. We also want to have growing boxes just for the kids so they can be in the garden war.

Level of food self-reliance: (present and future): diet and taste preferences

1) Existing: We have a beautiful garden right now. But we still rely a lot on the Farmers Market. We also go to Rainbow Bridge. Our food program cost \$225 and Everybody Eats for free for the whole month. It also includes dish soap and detergent and toilet paper. We buy just about everything in bulk.

2) Desired: we want to have a much bigger Garden. Maybe three times the size of our present one. We would like to have true food sovereignty. We'll need more Gardeners of course and we will need a better watering system.

Does income: need to be produced from the property?

1) Existing:

At this point income is not really necessary for the property. but we are working towards

getting income by selling food at the farmers markets. Most of the income comes from rent.

2) Desired:

Our garden will produce income for the farm. We want to sell things at the Farmers Market. Also we have a unique Workshop. We can be rented out. We also want to have a Repair Cafe. Also an apothecary would be nice. We could also be a demonstration site for permaculture .

Compost and recycling systems?

1) Existing:

Our food from the kitchen goes to the chickens right now. In the past we had a better composting system. Which involved three big mounds that we would turn every 3 days. It produced beautiful soil and it was a lot of work. recycling consists of a large container at the front of the property 4 cardboard plastic and glass.

2) Desired:

We want to have a better composting system that separates food for the chickens and food for the garden. We also want to make sure that we stay anaerobic with our compost pile. It would be great to have a recycling system where we could get money for metal and our bottles and aluminum cans.

Livestock:

1) Existing:

Right now we have about 40 chickens which produce about two and a half dozen eggs everyday. This is perfect. The grain is scratch and laying pellets that last about 2 weeks and cost \$100.

2) Desired:

We would like to have the things we had in the past. In the past we had three cows, a whole bunch of pigs, and three sets of goats, We also had sheep and ducks. What happened was that when key people left the farm we had to give up the animals because it was too much work.

Pets:

1) Existing:

We have three big dogs which bark way too much. And we have about 10 cats.

2) Desired:

I personally would like to have 0 dogs and 20 cats.

Wildlife enhancement:

1) Existing:

We have Bobcats, owls, red-tailed Hawks oh, lots of hummingbirds oh, lots of finches, too many gophers. We also have ground squirrels and rabbits

2) Desired: Less gophers and more bobcats.

Fencing:

1) Existing:

We just built a fence of chicken wire around our 1/2 acre Garden. It does need to have two more gates. We have put wooden fencing up in key places to shield us from the neighbors.

2) Desired:

We would love to have cob walls around our whole property, but this would probably be way too costly. We are right next to the road so having a sound barrier and a privacy barrier would be really nice.

Woodlot:

1) Existing:

There are a lot of trees on this property, mostly chinaberry and Atlantis trees. which which grow really fast oh, so we use it for a lot of firewood. We also go and get Oak from various places which is a really nice firewood.

2) Desired:

We actually need things to process like chainsaws and wood splitters and renewed wood storage places with roofs against the rain.

Vehicles: (car/truck, farm equipment, recreational equipment, earth moving equipment or other heavy machinery?)

1) Existing:

We have a Ford F-250 pickup which is our work horse and we just got a tractor with a backhoe, front bucket and a wood chipping attachment.

2) Desired:

With the purchase of the tractor we are set for a while.

Summary

What is the client's vision for the site?

Our vision as a community is to grow our own food, practice our own governance, be independent from society in general but interact in a positive way as a teaching facility for 2 other communities around. We want to focus primarily on permaculture and we all have a spiritual connection with nature and conduct lots of nature-based ceremonies which we would like to share with others. Our carrying capacity for the land is about 20 people, so we would like to have a Full Circle Farm 2, because we are in very high demand as a community. Lots of people want to live here. We want to be an awesome permaculture site!!

What are the challenges or limitations that the client has encountered on this site?

Some of the challenges as for the site is that we live in a very expensive area. Also we have challenges with self-governance. We have tried sociocracy, and holacracy. These are really hard to perpetuate here at the farm, as a lot of new people just don't understand how it works.

What are your goals as the designer for this project

My goal as the designer of this project is to inspire people in creating a very beautiful community that is self-sufficient, that is socially just, that is spiritually awake, where people interact as a family, using the resources at hand. I believe we've been working toward this end goal for 30 years. most of the work has been done in Prior years. I will simply be standing on the shoulders of giants

3.1. Site & Regional Challenges Survey

There are many ways to think about and implement permaculture, but ultimately the goal of permaculture is to think far past ourselves, into our descendants and their descendant's time. As ethical designers, we should ask ourselves, how does our design relate to the community that it is in? Does it respond to the challenges of the ecological and human communities around it?

You may expand on your answers by including images if you'd like.

- What are the three biggest cultural / social / economic challenges in your region?
Ojai is a culturally affluent area so the economic and social challenges are that things actually cost more than in other places. Also the cost of Labor is higher also.

- What are the greatest strengths or opportunities in your region?

Ojai has been beautiful farmland for a very long time. Also the people are health conscious and might be more inclined to use organic farming methods.

- What are the three biggest problems or weak links on your design site? (climate, water, access, vegetation, regulatory, interpersonal, etc)

Ojai is very hot in the summer, and we do not have a Dependable long-term water supply. City water comes from Casitas Lake and the price is going up as the lake water level goes down. There's also a lot of regulation in Ojai, but we do manage to fly under the radar.

- What are the greatest strengths or opportunities on your site?

The greatest strength is that we can attract good people to our site because the region has a lot to offer.

- What are the most likely natural disasters to happen in your region?

Fire is the most likely natural disaster. Every year we have to do a large amount of fire clearance. The Thomas Fire swept through this area and it was a disaster.

- What are some strategies for resilience against those natural disasters?

Fire clearance, along with planting fire resistant plants, and mitigating brush is the best strategy.

3.2. Sector Analysis

Remember, **sectors** are natural elements/influences/energies impacting the site that we have little control over. However, we can influence how these elements/energies move across our site, by **channeling them differently (aka redirecting/diverting them), opening a channel for them (e.g. wind-tunnel for cooling), or blocking the channel**. Remember to consider the constraints and opportunities as you do this sector analysis - which sector is the most influential? How will this affect your future site design? What opportunities present themselves despite the constraints?

In this assignment, consider the different sectors affecting your site, such as:

List/map each sector and name what direction it is coming from and what your design response will be: if you intend to channel, open or block it. You may also draw this if you'd like.

Sector Analysis Map



1. **Warm or Cold Wind:** Full Circle Farm is in a wind dead zone. when the wind forecast is 10 mph in Ojai oh, then it will be 4 miles an hour here. occasionally we will get some wind but not very much. Usually the wind is from the West, as you can see by the **big blue** arrow on the map. but when it's Santa Ana season the wind comes from the East. This is a really dry wind that creates fire danger and nobody likes it.

2. **Fire:** We are in a very fire prone area. Especially this Farm which is very overgrown. We are more in danger than most other Farms. A lot of the brush is down in the Zone 5 Lion Creek area, as depicted on the map by the red flames. The Thomas Fire came through here in 2017 and burned several buildings and dwellings but did not touch the main structures. It missed the house, the workshop and the barn. We are constantly doing fire clearance but we are always behind for lack of manpower. We have a brand-new chipper that will help with brush clearance.
3. **Shade:** You can see on the green on the map that about 50% of the property is under trees so there is a lot of shade on the farm. We consider the shade very valuable to the quality of life here.
4. **Summer and Winter Sun:** There is a good sun position calculator at https://www.sunearthtools.com/dp/tools/pos_sun.php?lang=en
5. **Water flow or flooding:** Every year our Creek Rises several feet. in the past it has come almost up to the house but this is a very rare occasion during a flash flood. It's only happened once or twice. It drains down pretty quickly. in the summer months it's completely dry. we have the potential of getting flooded but it's actually never happened.
6. **Noise:** we are very close to the main road so we get a lot of noise from traffic, particularly on the weekend when motorcycles come through here and we have a lot of motorcycle noise. It does die off at night time. Sometimes we hear our neighbors partying because they are pretty close. We hear a lot of dogs barking and coyotes at night.
7. **Views:** Because we are low down we actually don't have a view. However two neighbors in the north are above us and they can look down on us.
8. **Access:** We have a single access route that is poorly presented on Google Maps. There is a fire exit in the back that will take us to the neighbor's yard.
9. **Privacy:** We actually do not have as much privacy as we would like. We are visible from the road and from The neighbors, and since we have a lot of people living here it doesn't feel very private sometimes.
10. **Wildlife :** We do have a fair amount of wildlife in the Lion Creek corridor. We also have coyotes come onto the land from the Padres National Forest behind us. We have lots of different kinds of birds like red tailed hawks, owls, crows and turkey vultures.

To fill out your sector analysis further, please include:

Closest Grocery Store - Distance and Direction:

The Summit Convenience Store located two miles up the road. It has plumbing supplies

and fast food.

Closest Population Center - Distance, Direction, Population:

Although the Ojai Valley has a population of **nearly 30,000** people, the City of Ojai is only four square miles and has a population of 7,400 people. Ojai is 6 miles down hill to the center of town.

Closest Fuel Station - Distance and Direction:

The Shell station has the best gas price. It's 5 miles down towards ojai.

Nearby Stores - Distance, Direction, Type:

Rainbow Bridge Is the best store in the area. It's almost fully organic. The prices are a little bit higher but it is the community center for most of the Progressive people in Ojai. Westridge is also a little closer at 5 miles and has everything anyone would need. Vons is 8 Mi away and a lot of people like to shop there.

Closest Hospital - Distance and Direction:

The closest hospital is the Santa Paula hospital that's 10 miles down the road towards Santa Paula, our other main city. There's also a Clinicas in Ojai, also 10 miles away.

Closest Public Space - Distance and Direction:

We have Soule Park 4 miles away down the hill, where people have picnics and there's a playground for the kids. The other huge public space is the Padres National Forest which is a vast region right behind our community.

Closest Market - Distance and Direction:

Ojai has a farmers market every Sunday from 9 until 1 p.m. where there are many organic things. There's another Market in Ventura on Saturday morning and a very small Market in Santa Paula Thursday.

3.3. Current Zones Analysis

Remember, the **zones** help us organize our site by use. In permaculture design, zone 1 is the house or primary structure and the area directly around the house/primary structure, with zone 5 being the farthest away from the center of the site (our house in this example), and usually is an unmanaged wild area if your site is a rural property. Not all

1. **Zone 1** Is the heart of the community. probably the area around the house, the workshop, and the bath house. This is where most people cross paths. people often walk through the workshop to get to the bathrooms on the other side. but then people are constantly walking between the cabins visiting each other for whatever reason. so it's easy to find people in that area along the pathways. All of Zone 1 is pretty shady all year long. Most of the bathrooms and showers are in zone 1. You'll pretty much find somebody somewhere in Zone 1 at any given time
2. **Zone 2** is the garden and the orchards. We have a garden of about a half acre. And orchards that take up about 2 acres. We also have a large chicken coop with about 40 chickens that get fed daily and I let out in the evenings to run around and forage. The orchard of Persimmon Hill on the North has exclusively persimmon trees. The orchard to the West has a mixture of fuyu and hachiya persimmon trees, grapes, pomegranates, plum orange, lemon, peach and guava trees.
3. **Zone 3.** Has a lot of trees and open grass that dries and creates open spaces. This area we have to cut back with lawnmowers. Half of our orchards are also Zone 3 as they are more removed and less frequented, and so to take care of themselves.
4. **Zone 4** is the unused area primarily to the north of the Farm. It's the borderline between us and our neighbors. We just sort of let it over grow to create a visual barrier for my neighbors. It's where a future cob wall is going to be.
5. **Zone 5** Is really wild and is our creek wetlands. It's south of the property and basically comes right up to the house. at the house there's a steep drop-off into the creek bed. The area is completely overgrown and impossible to pass through without a machete. In the winter time the water rises sometimes quite High oh, that normally doesn't have flowing water through the creek for more than half the year. There's all kinds of wildlife down there. In the summer the area poses a real fire hazard, but generally it's too much work to deal with, so we just leave that whole area alone.

Identify all traffic flows & pathways. Name the type of traffic (foot, car, dog etc.) and the frequency of use (1x/day, 3x/week etc.) Identify all doors, gates, entries and exits.

The most common daily traffic patterns are to the kitchens and the bathrooms which are mostly in the house. people will do this multiple times a day. also the bathhouse gets

constant use. The workshop also gets constant use. The garden is used daily and the chickens are fed daily. The orchards are visited weekly for watering. People usually don't go into a zone for any reason except to cut grass. And the wetlands by the creek people almost never go.

Explain what occurs in each of your zones. Feel free to add images to support your explanations.

Zone 1 is the living area

Zone 2 is the vegetable garden

Zone 3 is the orchard.

Zone 4 is left alone

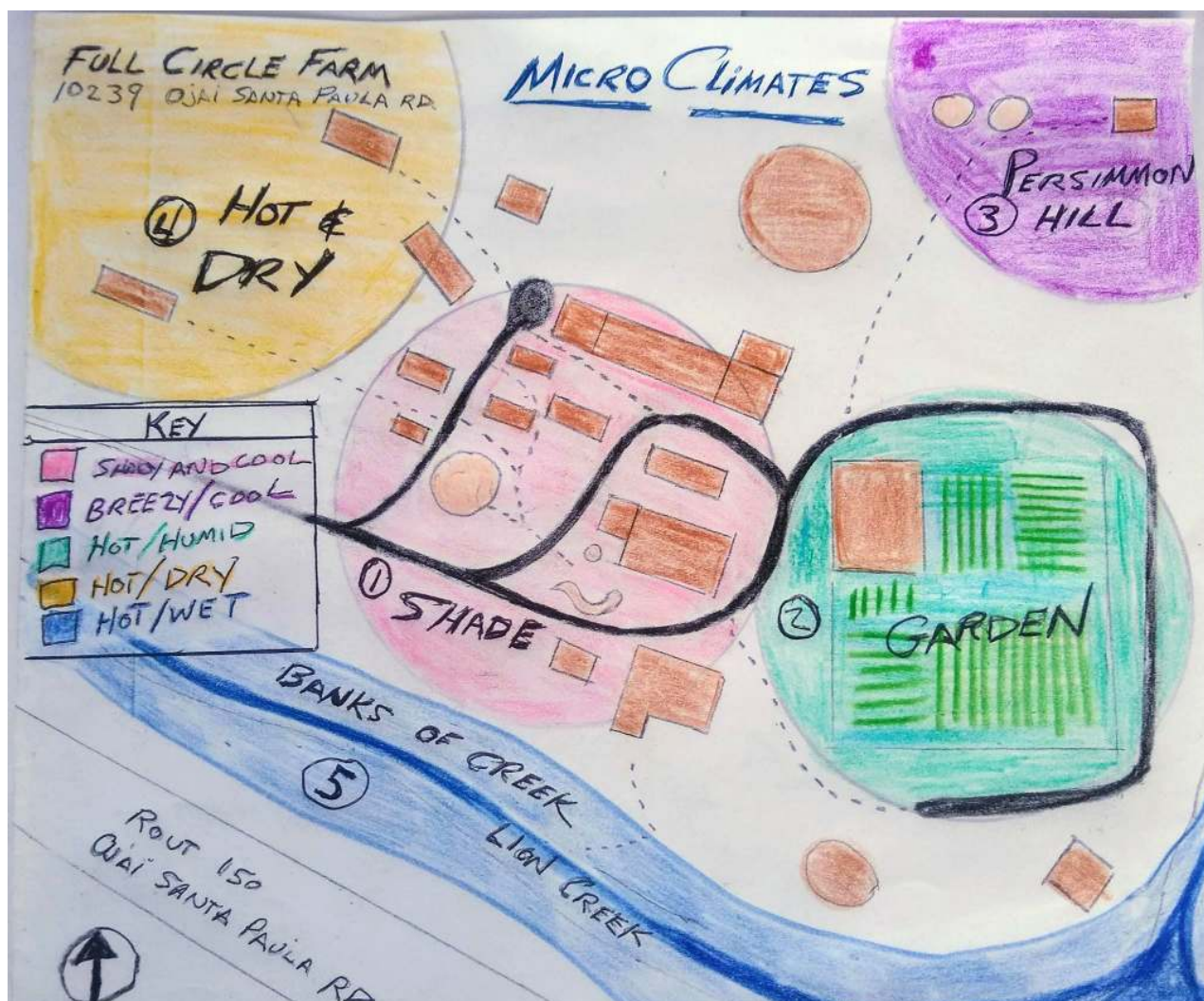
Zone 5 is the wildlands

3.4. Microclimates

What microclimates exist on your site? Where are these microclimates in relation to each other? If you are mapping, you may answer this question visually.

There are five basic microclimates at Full Circle Farm.

Microclimate Map



1. **Farm Center:** The shady area that makes up most of the structures on the farm. There is really good tree coverage creating cool shade. We have Atlantis, Chinaberry tree, evergreens, loquats and redwoods. The loquats in fact produce the most amount of shade.
2. **Vegetable Garden:** The garden is completely exposed to the sunshine oh, but it has a little more humidity because the garden sprinklers are on a lot in the hot weather. Sunflowers grow very well here.
3. **Persimmon Hill** has a higher elevation by about 25 feet and it gets a little more breeze. So that end area tends to be cooler, and breezier. The persimmon trees don't get watered as much as the garden so there's less trapped humidity. Also the eucalyptus trees that Tower over Persimmon Hill probably absorb a lot of water making it more dry.

4. **The Hot and Dry** northwest of the farm is also less-developed, it's the far end of the West Orchard and it feels pretty dry there usually. There are eucalyptus trees towering over this area but the sunlight still hits it year round.
5. **The Banks of Lion Creek** Are the wettest and shadiest areas. It's downright tropical in Lion Creek, which has steep Banks that go down 25 feet. During the winter the creek flows strongly and Rises 10 ft Maybe. In the winter it's cool and damp and there are a lot of frogs croaking all night long. In the summer the creek, particularly more recently, dries up but remains cooler, because cool air sinks.

Identify 5 plants on your site, each in a different microclimate. For each plant, name the family, genus, species, and common name. You may also include a photo if you'd like. For each plant please list (and please cite your sources):

- Native location and preferred habitat
- Microclimate (light, water, soil, exposure, temperature)
- Form (tree, shrub, herbaceous perennial/annual, grass)
- If it is edible, medicinal, useful, poisonous, etc.

Zone 1: Jasmine (taxonomic name: *Jasminum*; [*/ˈjæsmɪnəm/*](#), YASS-min-əm^[5]) is a genus of shrubs and vines in the olive family (Oleaceae). It contains around 200 species native to tropical and warm temperate regions of Eurasia, Africa, and Oceania. Jasmines are widely cultivated for the characteristic fragrance of their flowers.

Zone 2: Peppermint (*Mentha × piperita*) is a hybrid species of mint, a cross between watermint and spearmint.^[1] Indigenous to Europe and the Middle East,^[2] the plant is now widely spread and cultivated in many regions of the world.^[3] It is occasionally found in the wild with its parent species.^{[3][4]}

Zone 3: White Sage, *Salvia apiana*, bee sage, or sacred sage is an evergreen perennial shrub that is native to the southwestern United States and northwestern Mexico, found mainly in the coastal sage scrub habitat of Southern California and Baja California, on the western edges of the Mojave and Sonoran deserts.^[1]

Zone 4: California poppy, *Eschscholzia californica*, or golden poppy, California sunlight or cup of gold, is a species of flowering plant in the family Papaveraceae, native to the United States and Mexico. It is cultivated as an ornamental plant flowering in summer (spring in southern Australia), with showy cup-shaped flowers in brilliant shades of red, orange and yellow (occasionally pink and white). It is also used as food or a garnish. It became the official state flower of California in 1903.

Zone 5: Pacific Poison Oak,^[1] *Toxicodendron diversilobum* (syn. *Rhus diversiloba*), is a woody vine or shrub in the sumac family, Anacardiaceae. It is widely distributed in

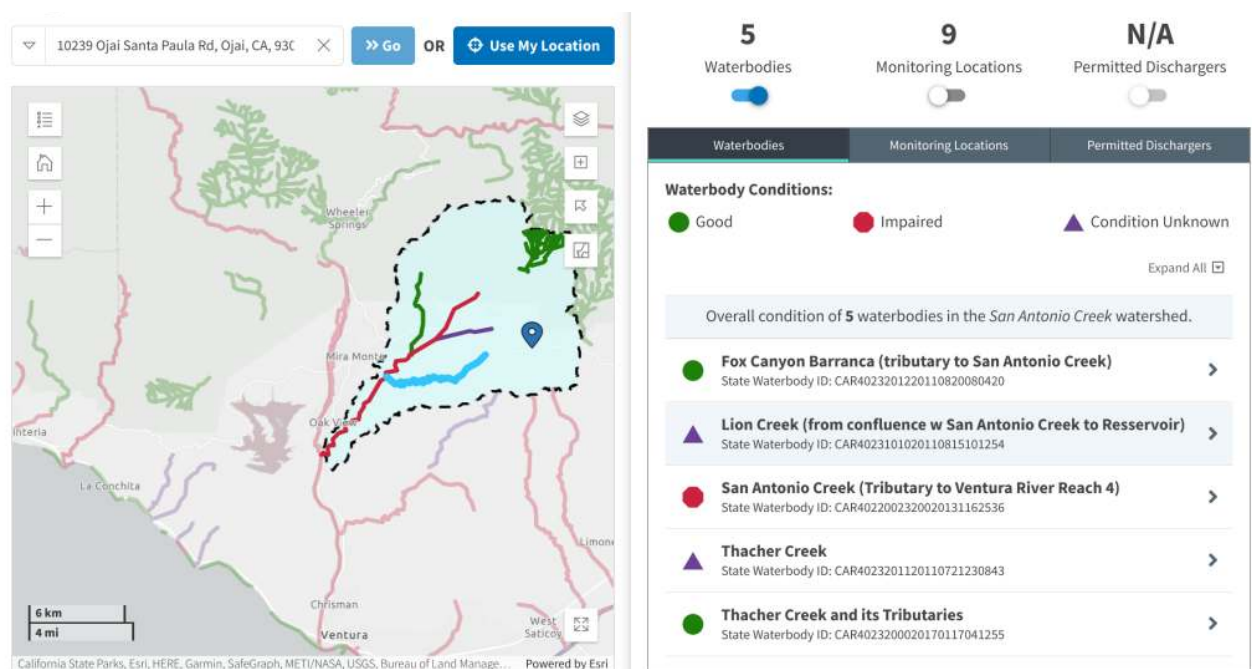
western North America, inhabiting conifer and mixed broadleaf forests, woodlands, grasslands, and chaparral biomes.^[2] Peak flowering occurs in May.^[3] Like other members of the genus *Toxicodendron*, *T. diversilobum* causes itching and allergic rashes in most people after contact by touch or smoke inhalation. Despite its name, it is not closely related to oaks.

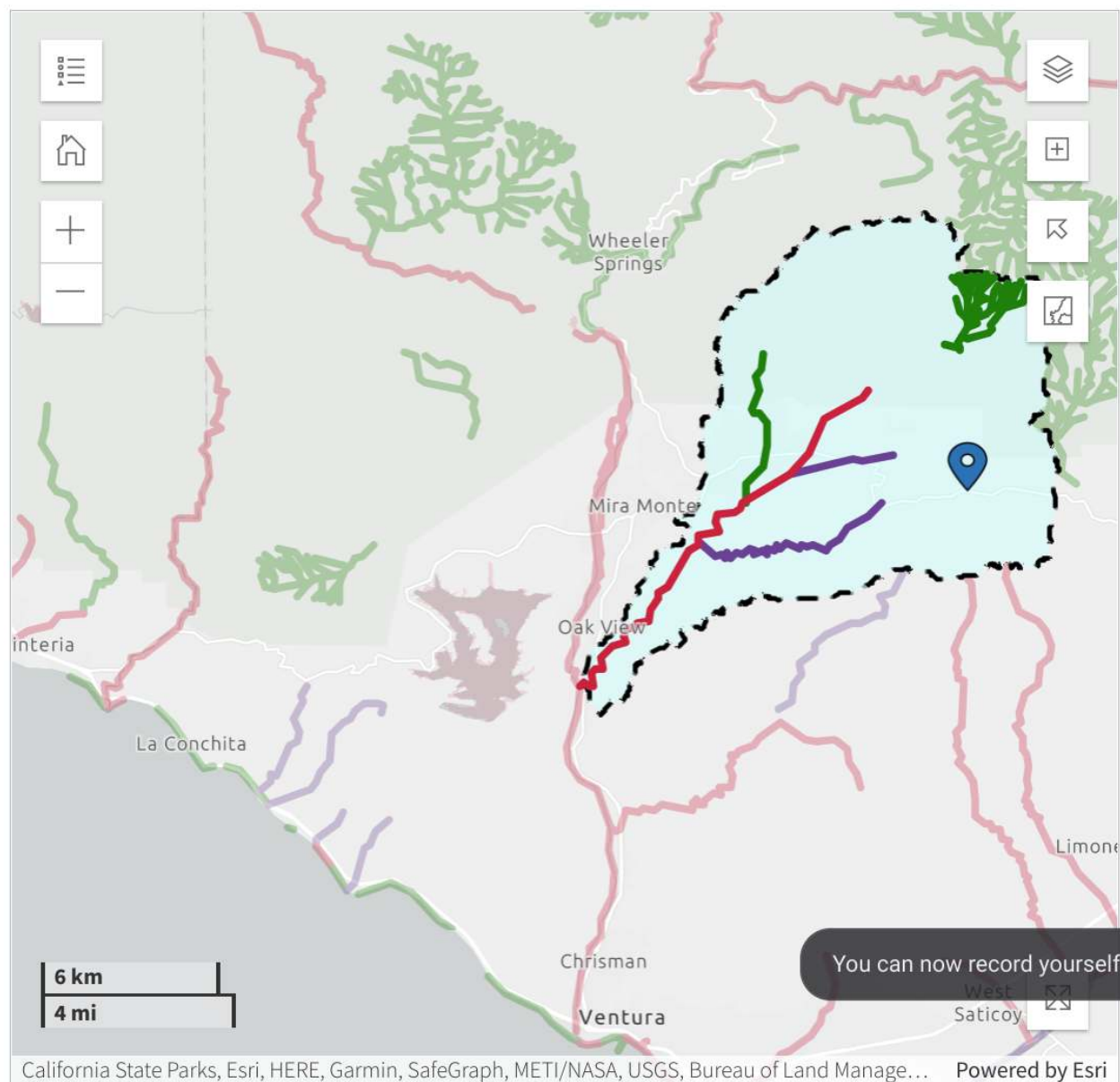
4. Water Survey

What is the watershed your site is in?

San Antonio Creek Watershed and tributary to Ventura River Reach 4

*US students can find their watershed through the EPA's [How's My Waterway](#) software.





Where does the water come from for the home/primary structure on your design site?

We have a 160 foot well which pumps about 20 gallons a minute.

We are also on city water. We have a pumping station right next to our property which supplies high pressure water through a 2 inch pipe. Recently we have turned off city water but may occasionally use it when we are watering and pressure from the well is low.

Where does the wastewater go?

Wastewater goes to several different places. Blackwater from the toilets goes into septic systems. We have two of those stops. Actually all the bathtub water and sink water goes into septic systems also, but I'm changing that out with a brain strain system. We have two composting toilets also. water from the washing machine gets dumped into the

flower bed, but I want to extend that line so that it reaches more beds. also I want to put tri valves on all the sink drains to water the nearby areas. our reverse osmosis water system drains into the flower bed.

What is the average precipitation per month, for each month? 20 incher a year.

January: 3
February: 3
March: 2
April: 2
May: 1
June: 0

July: 0
August: 0
September: 1
October: 2
November: 3
December: 3

What is the total area of your site? (length x width = ft²) OR (length x width = m²)

217,800 square feet

What is the yearly average rainfall (in feet or millimeters)?

1.6666 feet

Calculate the Total Rain falling on site in **average year** (length x width x volume) (in **gallons or liters**):

Imperial: Area (ft²) x Total Rain Avg (ft) x 7.48 (conversion to gallons) = total gallons

Metric: Area (m²) x Total Rain Avg (mm) = total liters

2.7 million gallons

What is the record 24hr rainfall for your design site (or closest place you can find)?

6 inches

Calculate the **total rain** falling on site in a **100 yr 24 hr event*** (in **gallons or liters**):

Imperial: Area (ft²) x Total 100 yr rain (ft) x 7.48 (conversion to gallons) = total gallons

Metric: Area (m²) x Total 100 yr rain (mm) = total liters

** What you are doing here is calculating (and designing) for a potential catastrophe - an 100yr/24hr hour event means a once in a lifetime event that could cause immense destruction (or opportunities depending on design!). In a time of extreme events, it is particularly important to know what an event like this could look like on your site.*

814,572 gallons

How is your site impacted in a major precipitation event and why?

Our property is relatively small, so rain will flow downhill where it will drain into the creek. Sometimes the Creek will rise and get very near the house, but in 30 years it has not reached the house. drainage is pretty good around the creek but not on compacted drivable surfaces and roads.

What are the roofing materials on your design site (if any)?

We use standard asphalt shingles on the house. The barn and workshop have tin roofs. We also have some polycarbonate roofs on the greenhouse and other small structures.

Calculate the potential water catchment of your roof (either part or all of it):

Imperial: Area (ft²) x Total Rain Avg (ft 2) x 7.48 (conversion to gallons) = total gallons

Metric: Area (m²) x Total Rain Avg (mm) = total liters

1000 square feet x .6 feet = 600 X 7.48 = 3,740 Gallons

4.1. Site Water Flow

Map or describe the water flow on your site.

Site Water Flow Map



If you are mapping, please use different colors or types of lines for each element and include a key.

1. **Elevation.** Full Circle Farm is between 1240 feet and 1300 feet. Most of the property is at 1355 feet. Contour lines are gray.
2. **Rain** falls across the property, and generally flows South towards Lion Creek.. The property is mostly clay. Water does not absorb quickly. Except for Persimmon Hill, and the banks of Lion Creek, the property is pretty flat. Water

will pool on the roads. We have a lot of vegetation which needs a lot of water so it all gets absorbed eventually. We have mulched extensively across the land, especially in the orchards and garden, so this helps with absorption.

3. **Well Water.** Our well is 160 feet deep and pumps 30 gallons a minute. This is our main water supply but water travels through a 2 inch pipe, which allows us to use no more than four hoses at a time.
4. **City Water.** We also have city water which comes in on the south east side of the property through a 2 inch pipe. We try not to use this water because it's expensive and it's chlorinated. Sometimes we are forced to use it because our well water is insufficient in the summer, especially when we are watering the garden.
5. **Roofs.** The largest roofs are on the house, workshop and barn but gutters just dumped into the ground next to the structures. One of our goals is to harvest Rainwater and store it.
6. **Impermeable surfaces** most of our land is 90 plus percent clay.
7. **Creek.** We have Lion Creek running through the property which is wet three quarters of the year. But it dries up in the summer.

4.2. Local Ecology Survey

In permaculture we use pattern understanding to first observe patterns in the natural world, and then do our best to apply those patterns best suited to a site as part of our site design. While you are conducting this ecological survey, think about the patterns you are observing in nature, and think about how you might later apply those patterns to your site design (i.e. replicating the native plant succession chain, or planting a medicinal garden that replicate some shapes you see occurring in your local ecology, etc.) We also want to remember one of the goals of permaculture, in line with the core ethics, is to be a source of regeneration on damaged landscapes.

When you are conducting this survey, think about whether or not these ecological communities have been affected adversely by current and expanding human settlement, climate change, industry, etc. Include this as part of your weekly reflection.

If the answer is yes, think about how your design will respond to these effects. Maybe you observe a set of plants that you notice birds and pollinators are frequenting - take note of that, do more research and decide whether or not you may include those plants in your final site design as a way to support the bird and pollinator populations that are being adversely affected in your area. Include this visioning and research as part of your weekly reflection.

Describe a local biological food chain.

We have gophers and rabbits on the land. These get eaten by hawks and owls. We have coyotes and bobcats on the land, and these sometimes eat our chickens.

Name five birds that inhabit your region.

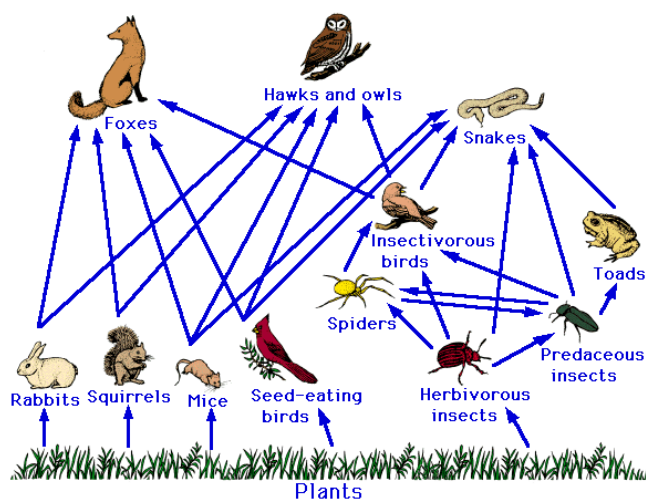
- *Red-tailed hawk*
- *Owl*
- *Hummingbird*
- *House finch*
- *Thrasher*

What is the name of a climax community of flora and fauna in your area (also known as biome, ecological community, ecological type)?

Climax community is an ecological community in which populations of plants or animals remain stable and exist in balance with each other and their environment. A climax community is the final stage of succession, remaining relatively unchanged until destroyed by an event such as fire or human interference.

California Chaparral Biome

Name five species and a food web that are present in your climax ecosystem.



Rabbits, Squirrels, Mice, Woodpeckers, Hawks, and Owls

Name five pioneer plants in your ecosystem.

A pioneer plant is a plant that occurs early in a vegetational succession. Pioneer species possess characters that suit them to their ecological niches, notably rapid growth, the production of copious, small, easily dispersed seed, and the ability to germinate and establish themselves on open sites.

Mustard
Horehound
Fiddleneck
Caterpillar Phacelia
Deerweed

4.3. Local Plant Survey

What are five native or naturalized medicinal plants in your area and what are they used for? What part of the plant is used?

From: <https://bloomcalifornia.org/>

Sweet grass is an aromatic, native perennial, with an extensive range in the United States, Canada and Eurasia. Growing in wetlands, prairies and meadows, it prefers

rich, moist, soils and at least half a day of sun. The grass flowers sit just above the foliage where they can dance with the wind. This cool season grass is sacred to indigenous people, where it is used in herbal medicine, a kind of tobacco, basket making, and burned as a smudge. Growing 1 - 2 ft. tall and spreading widely by sturdy rhizomes that can be difficult to eliminate once well established. Winter dormant in cold areas. Deer resistant

Hummingbird Sage is a herbaceous plant species with woody bases and a somewhat sprawling habit with upright flowering stems. It is a member of the large *Salvia* or sage genus in Lamiaceae, or the mint family. This fruity scented *Salvia* blooms in March to May with typically dark rose-lilac colored flowers. It is native to southern and central California found growing from sea level to 2,000 feet and is cultivated in gardens for its attractive flowering spikes and pleasant scent. It grows in the California coast ranges from the Sacramento Valley south to coastal Orange county in the south. A common species that grows on shady slopes in oak woodland, chaparral, and coastal sage scrub. It is commonly found in oak woodlands not far from the Pacific Ocean.

White sage, bee sage, or sacred sage: is a cornerstone species of the coastal sage scrub habitat of Southern California and Baja California. White sage is fragrant, with silver-white leaves, and clusters of white flowers with lavender streaks. Young leaves start off green and turn white as they get older. White sage is deeply rooted in the cultures and lifeways of indigenous communities of Southern California and northern Baja, the only region this sage naturally occurs in the world. It is also an important food source for bees, butterflies, birds, and other wildlife. Larger bees, notably carpenter bees, are the predominant pollinators of white sage. Wild white sage populations are currently under intense threat by development, smudge stick poaching, climate change, drought, and wildfire.

California Mugwort: *Artemisia douglasiana* is an aromatic shrub in the Asteraceae (Sunflower) family, although plants in this genus don't look like it. It is commonly called Douglas' Sagewort. Found in virtually every county of California, this plant has been used as a ceremonial plant in many Native American cultures, for use in treating a minor injury or as a dream potentiator. The plant is native to the western United States and into Baja California where it is often found in moist areas such as drainages. It's often riparian in the drier parts of its range. The flowers are small and inconspicuous, often not producing viable seed. It spreads by rhizomes, especially in damp areas. This plant is not often seen in residential gardens but is sometimes grown for medicinal purposes. It is also useful in restoration projects and for erosion control.

Yarrow: Achillea millefolium (other common names common yarrow, gordaldo, nosebleed plant, old man's pepper, devil's nettle, sanguinary, milfoil, soldier's woundwort, thousand-leaf, and thousand-seal) is a flowering plant in the family Asteraceae, native to the Northern Hemisphere. In Spanish-speaking New Mexico and southern Colorado, it is called plumajillo, or "little feather," for the shape of the leaves. In antiquity, yarrow was known as herbal militaris, for its use in staunching the flow of blood from wounds. Native Americans had many uses for the plant, including pain relief, fever reduction, and blood issues of all kinds. Yarrow grows up to 3500 meters above sea level. The plant commonly flowers from May through June, and is a frequent component in butterfly gardens. Common yarrow is frequently found in the mildly disturbed soil of grasslands and open forests. Active growth occurs in the spring.

Also:

Clover

Malva

Perslain

Nettles

Mullin

Have you used any of these or other plants for medicinal use?

I have used white sage, sweetgrass, and mugwort in ceremonies to clear space. I have used yarrow on a wound.

Name one wild plant food that people historically ate during each season.

Acorns

List the plant nurseries in your area.

- Meiners Oaks Nursery
- Flora Garden and Nursery
- Australian Native Plants

List some farms in your area that you would be comfortable purchasing your food from and why you would be comfortable purchasing from them.

- Farmer and the Cook in Meiners Oaks

- Beedee's farm in Ojai (has a farmer's market stand)
- Topa Vista Farm (just across the way from us)

4.4. Current Plant Systems

In this section you will map (written or visual) what plant systems already exist on your site. This assignment is meant to guide you to identify what planned and unplanned vegetation exists on your site, and spark your design thinking. Use your weekly reflection section to vision on how you will (or will not) include the existing plant systems in your final design.

Current Plant List

Right now we have the following trees:

Almond	3	Oak	23
Apple	3	Oleander	1
Apricot	16	Olive	14
Bamboo	1	Orange	7
Black Locus	6	Palm Tree	1
Black Walnut	8	Peach	1
Cedar	1	Pepper Tree	21
ChinaBerry	30	Persimmon Date	1
Cyprus	1	Persimmon Fuyu	5
Elderberry	10	Persimmon Hichya	10
Eucalyptus	12	Pine	25
Fake Cherry	4	Pineapple Guava	10
Fig	2	Plum	5
Goji Berry	2	Pomegranate	7
Grape	3	Poplar	1
Ivy	1	Redwood	10
Jasmine	3	Rosemary	3
Lemon	5	Rose Tree	3
Loquat	15	Tea Tree	1
Moneytree	1	Tree of Heaven	30
Mulberry	7	Willow	10
Nectarine	1		

Total number of trees 330

In your written narrative or visual map include everything that you can - identify hedges, garden beds, trees, herbs, native plants, etc.

We have over 320 trees on the property. It's not practical to actually create a map with this many trees. The advantage of so many trees is that there is a lot of shade everywhere. The disadvantage is that there's a lot of tree trimming and a lot of falling leaves that create fire hazards.

The largest number of trees are The Ailanthus (Tree Of Heaven) and the Chinaberry trees. Unfortunately these are invasive species and do not add to the quality of the soil. They don't make very good firewood either. Finally, they're not good for attracting wildlife. Ideally we would get rid of the Ailanthus and chinaBerry trees but this is just about impossible as the ailanthus trees are so well rooted and spring up virtually everywhere. The Chinaberries are equally difficult to get rid of.

We also have a large number of oak trees, pepper trees and pine trees, which is really good for this area. They are evergreen so protect from the sun in the wintertime which can also be hot and harsh.

We have 12 eucalyptus trees, and this is not a large number, but the trees are enormous. They drink up a huge amount of water from the ground. I've heard that eucalyptus that's not too long in this area and should be cut down. Their leaves are not good for mulch. The turkey vultures like to roost in them every night.

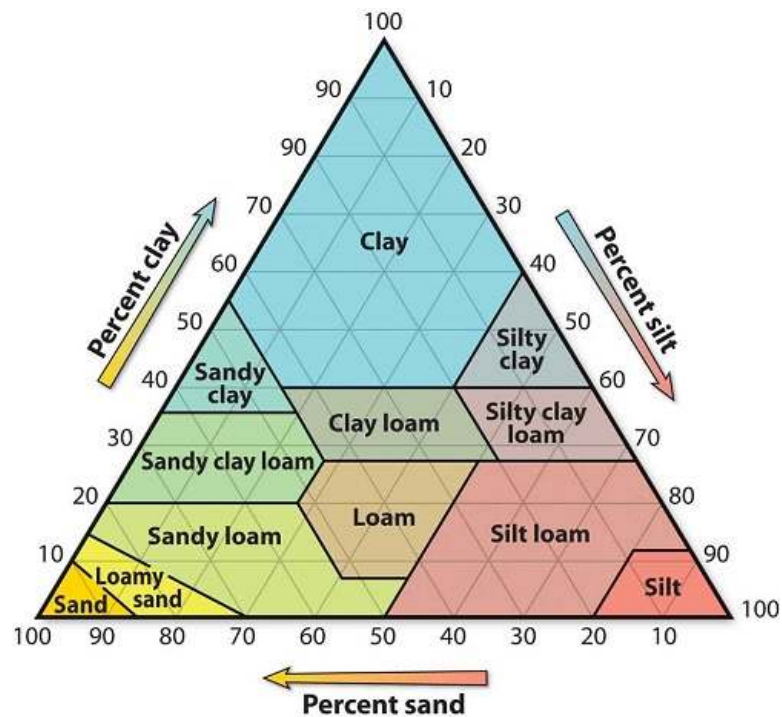
There is so much shrubbery on the land particularly in the creek Zone 5.

Every dwelling has at least a small garden in front or around it. We spend a lot of time watering these smaller gardens. We have 8 raised garden beds in the large vegetable garden.

Part 2

PDC weeks 13 - 16

5. Soil Tests



In this assignment you will be doing soil tests in 3 areas of your site. You can find links to instructions for these tests in your Design Project Flow tab for this week.

Jar Test:

Site 1

Percentage clay: 93

Percentage silt: 6

Percentage sand: 1



Site 2

Percentage clay: 80

Percentage silt: 19

Percentage sand: 1

**Site 3**

Percentage clay: 80

Percentage silt: 18

Percentage sand: 2



Texture by Feel:

Site 1 soil type: Clay

Site 2 soil type: Clay

Site 3 soil type: Clay

Percolation Test:

Site 1 drainage rate: 25 minutes

Site 2 drainage rate: 30 minutes

Site 3 drainage rate: 15 minutes



What 3 areas did you choose and why?

All three sites were in the garden. One was at the South End, One was at the North End and one was at the East End.

How do the different areas compare with one another?

All the areas were pretty much the same. We have most likely on the land and we have been amending it with lots of mulch fertilizer oh, but you can see floating on the top have the jars

What are the constraints and opportunities of your soil?

The soil could use a lot more sand. Otherwise it's pretty rich for planting.

What local materials can you use for a soil building plan that can minimize your limitations? We have plenty of compost from my kitchen oh, and we have plenty of mulch from chipping trees, and dead branches. Also, last year's crop gets mulched every year.

5.1. Regional Buildings Survey

What local materials presently or historically do indigenous people use to construct their homes?

The Chumash used a lot of willow or adobe traditionally. Today it's mostly stick and frame, like everything else. Cob, or Earthdome, or earthship (recycled materials) would be a possibility.

What can we learn from the traditions on this land for modern design?

The traditional designs used in a modern context are tried-and-true tested. They are appropriate for the region and the availability of materials. That stood the test of time.

Which driving force in your region did traditional materials address and how? What opportunities did using traditional materials provide?

Using traditional designs is better because it's simply just more available. We want to work with what's around us.

Are there current constraints (limitations/challenges) that using these traditional materials would help in your region/on your site? (e.g. due to current catastrophic wildfire events in California, one could say that traditional earthen building materials would help homes be more more fire resilient)

Some of the constraints include the building code which doesn't take into account natural materials very well, and the dominance of the timber and cement industries.

6. Zone 1 Design

Let's get designing! Over the last weeks you've worked hard to complete a number of different surveys and analyses of your site, meant to guide you towards a deeper comprehensive knowledge of your site.

Now it is time to put that knowledge into action throughout the last three weeks of the course by creating your own designs for the site based on the comprehensive knowledge you now have of your site.

To have comprehensive and informed knowledge of your site means that although you might have dreamed of putting in an almond orchard on one portion of the site, through the last six weeks of analyses and surveys you now realize that there is not enough rainfall nor a healthy enough groundwater basin to support that water intensive of a system on your site - you will now choose a different system because of that information you have. Permaculture design is not simply dreaming - it is an applied science that uses observation, visioning, assessment, design, and implementation. Now is the time to put those all into motion (well maybe not implementation just yet).

You're going to start this design process by creating your zone 1 map (visual or written). Remember, in permaculture we define zone 1 as the area with the most use on our site. If the center of your site is a house, this means zone 1 is the area immediately surrounding it. If the center of your site is a community center, then everything that surrounds it is zone 1. Refer to your zone 1 analysis from week 4 and any visioning you did around what elements and systems you would keep or transform.

Remember, zone 1 should truly be the closest possible area to your central dwelling. Make sure to review the *Design Methodologies 2* lesson section on zones for a refresher on what sorts of systems and elements might go in zones 1-5.

You are encouraged and welcome to include a visual map of your zone 1 design.

These are some systems/elements your zone 1 design might include:

- A kitchen garden and other plant elements
- A greywater system from the central dwelling

- A compost system
- Pathways and fences
- Parking area
- Patio/deck
- Community gathering space (i.e. outdoor kitchen, picnic tables, etc.)
- Studio or workshop
- Small animal systems
- Rainwater catchment systems
- etc.

Map or describe a detailed plan for Zone 1 of your design.

Zone 1 at Full Circle Farm is large because it encompasses the heart of our community. It's not just the house but the cabin's surrounding the house, the workshop, barn community rooms, showers, bathrooms, kiva and meeting spaces. Also our Zone 1 is very full of vegetation. Zone 1 is shaded by a lot of trees. Because there are lots of structures there are lots of roofs, but they are small.

We will make some important changes to Zone 1, but a lot will stay pretty much the same, as it has been in constant development for decades.

These changes I will describe in writing with a key that can be found on the maps included. So the Zone 1 plan will be both mapped and described in writing.

Zone 1 Design Map



1. **House.** This is really the heart of our community. We have two kitchens where people eat together. One Vegetarian inside, and one meat kitchen outside. The indoor kitchen dumps water into a garden bed. I will add a branch drain system to this to extend the range of this pipe. The outdoor kitchen dumps into the septic system. I will leave this the way it is. Please refer to the Final Water Map for details.

The house has a shower and 3 toilets. Diverting the shower water to the garden will be difficult as the piping runs under the house. The Toilet water will go to the septic, as always.

Four people live in the house, and we have a community room for meeting where we meet on a weekly basis for community meetings, conflict resolution meetings and heart circles. The roof will have rain water catchment. See the Final Water Map for details.

2. **The Workshop** is directly across from the house. We spend a lot of time there fixing things around the farm. Because it has a large roof we are going to install **solar panels** on the roof. We have been reinforcing all the calls and roof supports for 30 solar panels that will convert the farm from grid power to solar power, and save us a lot of money, and so we can be independent from the city.
3. **The New Meeting Room.** is the west side of the workshop has been unfinished for 30 years, and this year we decided to put in a cement floor and doors and a stage in the front for events. This has always been our dance spot for parties so it gets a lot of use. We are three quarters of the way finished with the renovation and need to get the walls finished and plastered. This will also be our new place for community meetings and other events.
4. **The Bath House** is also frequently used by everybody. We have a shower on the west side and a bathroom, with a bathtub on the east side. On the south side we have washing machines and dryers. We also have clothes lines which always have clothes on them. I will install a branch drain system that will take the shower water and sink water to the garden beds surrounding the bathhouse. There is a large Arrundo patch where the water is dumped. This will be extended to flower beds around the cabins to the West. see Final Water Map for details.
5. **Kiva.** This is one of the favorite spots on the farm where we have drumming circles and sometings meetings. It's our sacred space. We used to have tall

Ailanthus altissima (Tree of Heaven) trees surrounding the kiva providing shade but they died and were cut down so we lack shade over the kiva. We need to plant more red wood trees there, which will take time to grow, but we need some sort of shade structure too such as shade cloth and a structure to support it.

6. **Barn.** The Barn has been around since the beginning 40 years ago. It used to house mules many years ago. It's used for storage right now. It has no plumbing. It will be renovated into a finished space with a poured (cob) floor and used as a community art and office space. It has a large roof which will be used for rainwater catchment. I will place two 2500 gallon tanks on either side of the Barn. This is a later project 3 years down the road as we are working on the workshop area right now. On either side of the barn are dwellings which are well maintained and bring rent money to the community.
7. **Cabins.** Within Zone 1 there are 9 dwellings, each of which produces graywater that can be diverted to the local garden. Most of these dwellings are very small. They include two trailers, two motorhomes, seven cabins and a geodesic dome. We have six more dwellings that are in Zone 2 and 3. These are a tiny home, airstream trailer, earth dome, a FEMA tent and a bell tent for guests.
8. **Compost systems** are not on this map. There is a compost pile in Zone 2, the garden. It is neglected and needs to be turned and watered more often.
9. **Pathways and fences.** Pathways are brown on this map. There are no fences in Zone 1

Zone 1 Soil Fertility

If you plan to continuously plant kitchen crops in your zone 1 garden systems, how do you plan to manage the soil fertility?

We know that growing plants requires nutrients from the soil, but that resource must be replenished or we run the risk of degrading our soil. Alternatively, if you are starting with soil that is significantly degraded, what steps do you plan to take to support it back to a healthy level for planting, water retention, etc?

Answer the questions below to complete the assignment and create your zone 1 soil fertility plan.

How will you manage fertility in Zone 1 throughout the seasons? (*Answer this question in the categories below*)

Site prep prior to planting:

We keep the soil fertile in three different ways: compost, mulch and chop and drop. This requires soil testing too for our very clay soil.

- 1. We generate a lot of compost from our kitchen which feeds 20 people everyday. This amounts to two 5-gallon buckets of compost daily. A lot of this goes to the chickens. and we need to do more to send compost to the compost pile to generate rich soil. We did this in the past, and we'll do it again in the future. This is our Zone 1 Plan.*
- 2. In Zone 2 which is primarily a half acre Garden, we mulch after summer. We get a lot of wood chips from the neighbors and from tree trimming companies. We end up with three or four truck loads every year. Finally we chop and drop after every harvest.*
- 3. In Zone 3 which is primarily our 2 acres of Orchards. We mulch around the trees, and we add worm tea from our worm bin.*

Post-planting fertility management **Spring:**

In the spring we plant cover crops to keep the land from drying out. With mulch all the bare spots about 6 inches.

Post-planting fertility management **Summer:**

Post-planting fertility management **Fall:**

Cover crops

Post-planting fertility management **Winter:**

(insert writing here)

7. Local Network Survey

Research and list any permaculture groups/projects, grassroots community self-sufficiency projects, or other like-minded groups in your area.

Full Circle Farm network

Rainbow Ranch

Ojai Foundation

Rainbow Bridge

Quail Springs

Ecotopia Hot Springs

List any instructors, mentors, practitioners, design/build companies, interesting nurseries, and what they teach, build or grow in relation to regenerative living/agriculture or permaculture.

Cob Cottage Company: One of the pioneers in cob building. A nationwide inspiration

Cal Earth: earthbag construction using rice bags coiled into domes.

Ringling Cedars of Russia and the Kin's Domains concept that is like permaculture

Geoff Lawton Permaculture

Are there any sites of permaculture interest that are available for you to visit in your area?

Quail Springs is just over the hill.

East End Eden

SBCC Permaculture Site

Rainbow Ranch

Totem Ranch (biodynamic)

Sustainable Ojai

U4EA Ranch Enterprises

8. Final Design Vision

Mission Statement & Goals

What is the main purpose of your site?

Full Circle Farm Vision:

A Community that provides a supportive peaceful environment for people to consciously live in abundance and harmony with nature and each other.

Full Circle Farm Mission:

Using organic regenerative gardening, art, education and open hearted communication members are encouraged to use their unique skills to create a thriving harmonious community.

What are the goals?

Our goals are to be a self-sufficient community with food, political, economic and sovereignty. Be happy and garden.

8.1. Final Water Map

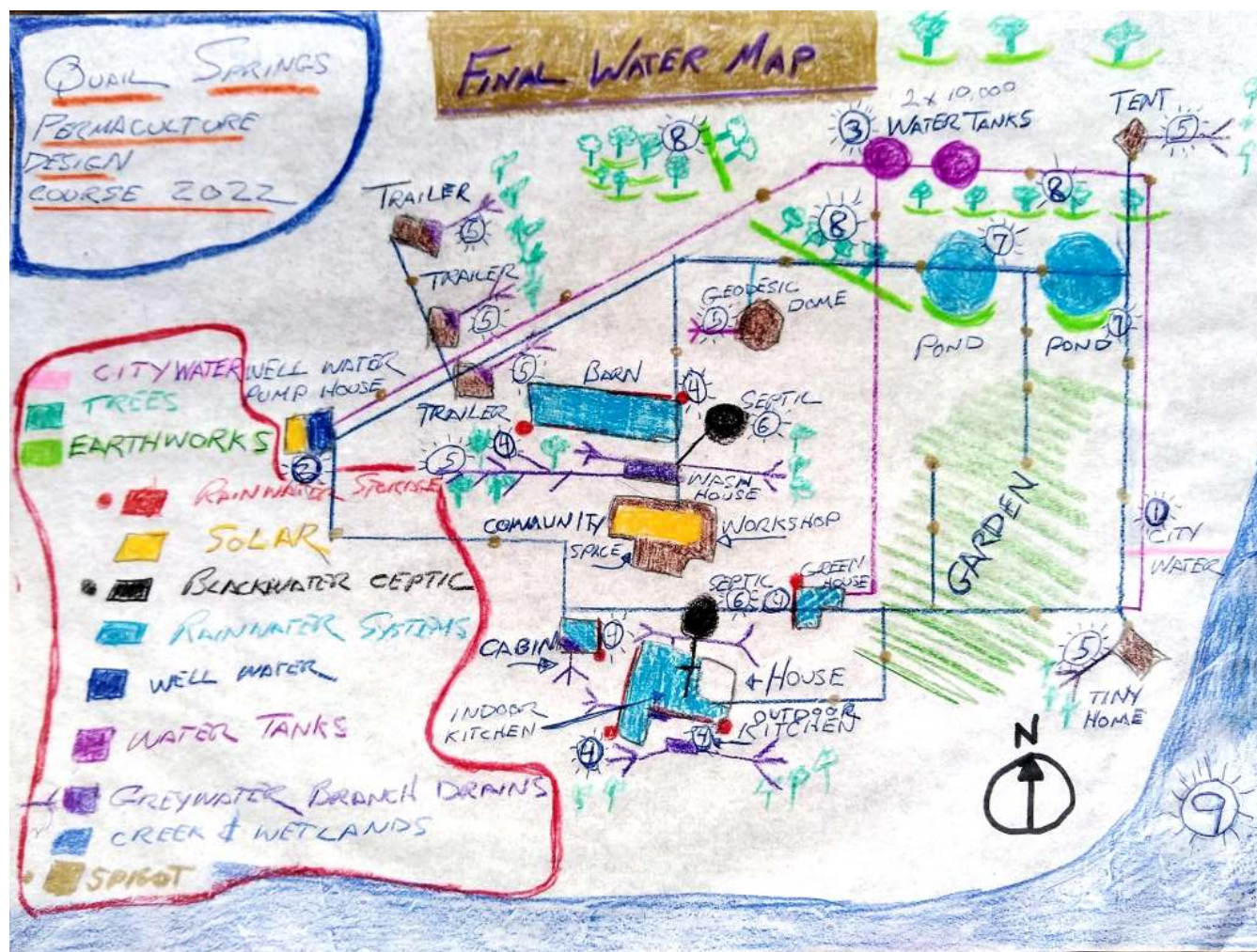
Create a written* or a visual water map for the site. This includes any new water systems on the site as well as designs to slow, use, divert or capture water for the entire site. Please account for all water entering, falling or being used on site. Please describe and/or illustrate the direction of water flow, especially in relation to other elements or systems on the site.

Some examples of systems or elements you may end up including in your water map could be:

- Rainwater catchment systems incl. rooftops and other methods
- Greywater systems
- Earthworks for rainwater harvesting or diversion (swales, berms, drains, etc.)
- Ponds
- Wetland areas

Full circle Farm is a reasonably developed permaculture site already. We have three water pathways presently and this plan will add two more. Look for the corresponding number (1-9) on the map and in the description below.

Final Water Map



1. **City Water:** We have a city water system That comes through a 2-inch pipe directly from the pumping station next to the property. Usually we keep the city water off because it's very expensive and getting more expensive. We can turn it on if we really need it to order something if it's really dry or if there's a fire or something happens to a well pump. City water uses the same water as well water and water from the water tanks.
2. **Well Water:** We have a well 160 ft deep and it pumps 60 gallons a minute through a 1 inch pipe. This is our main water source for the whole Farm and 1 inch pipe runs pretty much everywhere throughout the property. This is blue on the map . The main use of this water goes to the garden, which is also to the kitchens (indoor and outdoor) and the Wash House for the showers and washing machines.
3. **Water Tanks:** We also have two 10,000 gallon water tanks on the hill to the north. actually we don't use these much. It's really designed as emergency storage. They were useful before the well was dug but often somebody leaves a spigot on

somewhere and the tanks just drain, so we waste a lot of water. I just keep them shut off but they're still full. if we didn't have any water from The City we could store it in the tanks. Unfortunately the tanks don't have screens to prevent insects and bacteria from entering. Actually the last time we used them to supply water to the house half the people got sick because it's untreated water.

4. **Rainwater Catchment:** We want to collect rainwater off of 4 roof surfaces. Our Roofs are actually not that large. The house has about a thousand foot square roof as does the barn. We will place two 2500 gallon tanks at the house and barn to catch rainwater. We have a cabin and the greenhouse also has good roofs for rain catchment if they are even smaller so they will just get 1000 gallon tanks.
5. **Greywater:** Our dwellings consist of cabins, trailers, tiny homes, tents, a geodesic dome, an earth dome and several motor homes. Most of these have water coming to them from a spigot, usually a makeshift kitchen. And this greywater water gets returned back to the land. However we can do much better by using a branch drain system that will reach more flower beds right now. Right now the water gets dumped right on the ground. We have two washing machines. They can actually pump water quite a distance so that's when the chicken is longer and feeds a much larger patch of trees. We have a lot of persimmon trees that need constant watering, and each dwelling has a small garden next to it.
6. **Septic:** We have two septic tanks. much of the sink water and all of the toilet water and shower water presently go into the septic system. We will use the branch drain system to Water Gardens with shower water and sink water.
7. **Ponds:** A project started 3 years ago was to create 2 artificial ponds. We got an earth digger to dig out two large holes, and then the work stopped. In this design we'll finish the ponds and fill them with water from the well. Ideally we would like to fill them with rain water but this area just doesn't get enough rain. There are a lot of trees surrounding the two big holes right now but We want to have more fruit trees around them. This will supply water to the trees and prevent evaporation. One of the costs is getting the pond liner which is very expensive.
8. **Swails:** 8 years ago the major Earthworks were done to the land on the hillside of Persimmon Hill alerts where it was created to trap water running downhill from The Neighbors and sink it into the ground. It was successful and the persimmon trees on the hill happened to be thriving. smaller eyebrow swailes can be created around a lot of the trees.
9. The creek at the South east end of the property is zone 5 wetlands. This area floods during very heavy rains. The banks rise over 10 ft. I don't think there's any real way to catch this water. It's where all the runoff from heavy rains ends up. Otherwise the area is full of greenery, big trees, moist soil, lots of poison oak and nobody can really get down there because it's just so dense.

Part 3

PDC Weeks 16-24

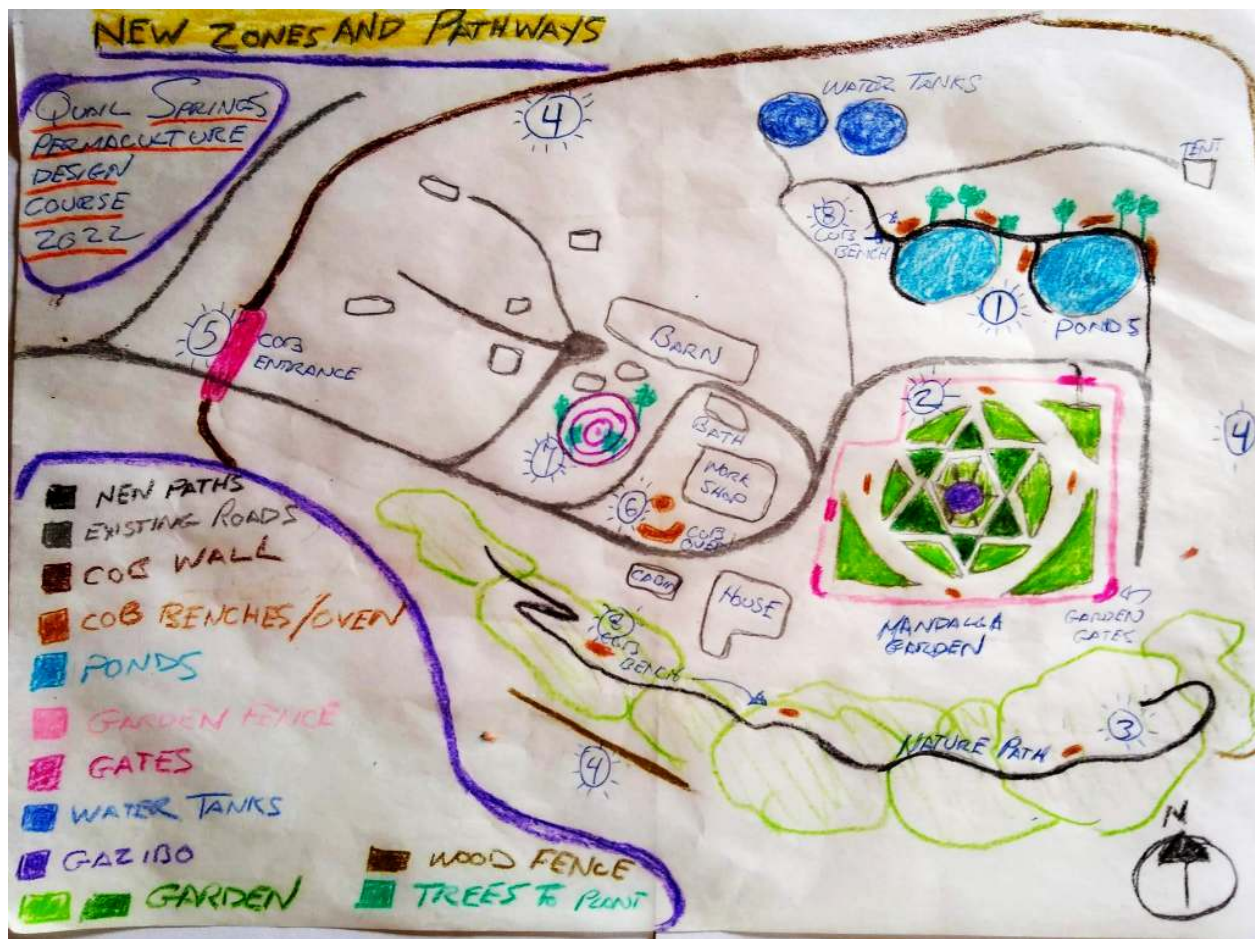
9. Final Design

9.1. New Zones and Pathways Map

Create a written or visual map of your site for any new zones and pathways. Some of the zones and pathways may remain the same as the original ones, but there should also be new zones and pathways that reflect your learning and design. Identify all doors, gates, entries and exits.

We are creating 3 new zones, cob walls and cob benches within the zones.

New Zones and Pathways Map



1. **Ponds:** The First Zone is around the two ponds. This will be a nicely shaded area with trees to prevent water evaporation. We can have tilapia fish in the pond. and The pond is also going to be surrounded by cob benches so people can relax.
2. **Garden Mandala:** The Second Zone is to redesign the garden into a Mandela star-shaped Garden. We'll use the six pointed star, and at the very center we'll build a gazebo for shade and relaxation. Right now our garden has no shade making it difficult to enjoy. The garden it's surrounded by a chicken wire fence which we started to put in this year. We need to finish the gates. This keeps the chickens out, along with the Wildlife living in the area.
3. **Wild Nature Path:** The Third Zone is the wild area nature path that goes from the heart of the wild area to the house. This is a beautiful area but nobody ever sees it because of the dense brush. By cutting a trail people can get down to where the creek is. It's cool and there's water down there. We can put the 3 cob benches down there too.
4. **Cob Wall:** The property is only five and a half acres and our neighbors are very close. Also the main road is very close, busy and noisy. Therefore we need a cob wall on the Northern side of the property. This will take a long time but it will help to find the space much better and increase our privacy. on the Southside And the east side we will put a wood fence. would with faster to build with then table
5. **Cob Entrance:** Most of the properties in Ojai have a large entrance way with a gate. We will build one also, out of cob that reflects the spirit of permaculture.
6. **Cob Oven and Bench:** We built a cob bench A few years ago, which has gotten a lot of use for Community meetings and parties. This year I started to build a cob oven which remains unfinished.
7. **Kiva:** The other space that gets a lot of use is the Kiva. It's been around for 20 years. All the trees around it have died so we need to plant shade trees for the summer.
8. **Cob Benches:** I placed 13 cob benches on the design. This is a lot of work so all the benches don't have to go in at the same time, however if we do offer cob building workshops then these are the places we can put them.

9.2. Final Vegetation Map

Map and describe the plant selections for your final design.
Explain your choices.

Please include:

- The mature size of your plant selections (how big will they get and how are you planning for that?)
- What type of human access ways there will be for plant care (i.e. where are the walking paths/roads/tractor paths/etc.?)

Final Vegetation Map



Because we have over 330 trees on the property, this map reflects new trees that will go

in over the next five years. Our goal is to remove all of the arundo from the farm, as it is highly invasive. Instead of arundo we will have bamboo. Also we have too much Ailanthus which grows like crazy. We have almost finished chopping all of it down. In its place we would like to have Paulownia, (which is somewhat less invasive but also quick growing). At one time Ailanthus was the most abundant tree on the farm. With great expense we have removed all the dangerous large trees. Another tree we wish to remove is all the eucalyptus and replace it with native sycamore. We have a lot of turkey vultures roosting in the eucalyptus. We will phase them out as we encourage tall sycamore. Eventually all the chinaberry trees will be removed as they are not good for the soil and surrounding plants.

We love avocados but in this microclimate only the Mexicola Avocados grow. They are frost tolerant to about 20 degrees. Black Locust also grows very quickly here and makes good wood for fence posts. Ceanothus hedge is very popular with the butterflies and bees so we want more of that, particularly to shield us from the roadway.

Oaks are already abundant and we will plant more oak. Figs grow well if the gophers don't get them first. Acacia is a large shade giving tree, and we love shade.

1. **Acacias** are graceful trees that grow in warm climates such as Hawaii, Mexico, and the southwestern United States. The foliage is typically bright green or bluish green and the small blooms may be creamy white, pale yellow, or bright yellow. Acacia may be evergreen or deciduous. Acacia Tree Facts Most acacia tree types are fast growers, but they usually live only 20 to 30 years. Many varieties are valued for their long roots which help stabilize the soil in areas threatened by erosion. The sturdy roots reach deep for underground water, which explains why the tree tolerates extreme drought conditions.
Acacia Tree Growing Conditions Acacia requires full sunlight and grows in nearly any type of soil, including sand, clay, or soil that is highly alkaline or acidic. Although acacia prefers well-drained soil, it tolerates muddy soil for short periods of time. Acacia Tree Care Acacia is basically a plant-it-and-forget-it type of tree, although a young tree may need protection from wildlife while it develops its defense system.
2. The **fig** is the edible fruit of *Ficus carica*, a species of small tree in the flowering plant family Moraceae. Native to the Mediterranean and western Asia, it has been cultivated since ancient times and is now widely grown throughout the world, both for its fruit and as an ornamental plant.[3][4] *Ficus carica* is the type species of the genus *Ficus*, containing over 800 tropical and subtropical plant species. A

fig plant is a small deciduous tree or large shrub growing up to 7–10 m (23–33 ft) tall, with smooth white bark. Its large leaves have three to five deep lobes. Its fruit (referred to as syconium, a type of multiple fruit) is tear-shaped, 3–5 cm (1–2 in) long, with a green skin that may ripen toward purple or brown, and sweet soft reddish flesh containing numerous crunchy seeds. The milky sap of the green parts is an irritant to human skin. In the Northern Hemisphere, fresh figs are in season from late summer to early autumn. They tolerate moderate seasonal frost and can be grown even in hot-summer continental climates.

3. An **oak** is a tree or shrub in the genus *Quercus* of the beech family, Fagaceae. There are approximately 500 extant species of oaks. The common name "oak" also appears in the names of species in related genera, notably *Lithocarpus* (stone oaks), as well as in those of unrelated species such as *Grevillea robusta* (silky oaks) and the Casuarinaceae (she-oaks). The genus *Quercus* is native to the Northern Hemisphere, and includes deciduous and evergreen species extending from cool temperate to tropical latitudes in the Americas, Asia, Europe, and North Africa. North America has the largest number of oak species, with approximately 160 species in Mexico of which 109 are endemic and about 90 in the United States. The second greatest area of oak diversity is China, with approximately 100 species.
4. **Sycamore**. This large tree grows to 35 meters (110 ft) in height with a trunk diameter of up to one meter (three feet). A specimen on the campus of Stanford University has a trunk girth (circumference) of 10.5 feet (3.20 m).[4] The trunk generally divides into two or more large trunks splitting into many branches. The bark is an attractive patchwork of white, tawny beige, pinkish gray, and pale brown, with older bark becoming darker and peeling away. *Platanus racemosa* is the dominant species in the globally and state endangered sycamore-alluvial woodland habitat.
5. **Paulownia** is a genus of seven to 17 species of hardwood tree (depending on taxonomic authority) in the family Paulowniaceae, the order Lamiales. They are present in much of China, south to northern Laos and Vietnam and are long cultivated elsewhere in eastern Asia, notably in Japan and Korea.
6. **Ceanothus** is a genus of about 50–60 species of nitrogen-fixing shrubs and small trees in the buckthorn family. Common names for members of this genus are buckbrush, California lilac, soap bush, or just ceanothus. "Ceanothus" comes from Ancient Greek, which was applied by Theophrastus (371–287 BC) to an Old World plant believed to be *Cirsium arvense*.
7. **Bamboos** are a diverse group of evergreen perennial flowering plants making up the subfamily Bambusoideae of the grass family Poaceae. Giant bamboos are the largest members of the grass family. The origin of the word "bamboo" is

uncertain, but it probably comes from the Dutch or Portuguese language, which originally borrowed it from Malay or Kannada. In bamboo, as in other grasses, the internodal regions of the stem are usually hollow and the vascular bundles in the cross-section are scattered throughout the stem instead of in a cylindrical arrangement. The dicotyledonous woody xylem is also absent. The absence of secondary growth wood

8. **Mexicola avocados** are a very popular cold-hardy avocado variety, a particularly hardy, vigorous species from the northern Mexican highlands. Plant one of these beauties and you'll have a bounty of 30 pounds of delicious avocados annually in just a few years.
9. **Black locust** reaches a typical height of 12–30 meters (40–100 feet) with a diameter of 0.61–1.22 m (2–4 ft). It is a very upright tree with a straight trunk and narrow crown that grows scraggly with age. The dark blue-green compound leaves with a contrasting lighter underside give this tree a beautiful appearance in the wind and contribute to its grace. Black locust is a shade-intolerant species[8] and therefore is typical of young woodlands and disturbed areas where sunlight is plentiful and the soil is dry. In this sense, black locust can be considered a weed tree. It also spreads by underground shoots or suckers, which contributes to the weedy character of this species. Young trees are often spiny, but mature trees often lack spines. In the early summer black locust flowers; the flowers are large and appear in large, intensely fragrant clusters reminiscent of orange blossoms. The leaflets fold together in wet weather and at night (nyctinasty), as some change of position at night is a habit of the entire leguminous family.
Although similar in general appearance to the honey locust, the black locust lacks that tree's characteristic long branched thorns on the trunk, having instead pairs of short prickles at the base of each leaf; the leaflets are also much broader than honey locust. It may resemble *Styphnolobium japonicum*, which has smaller flower spikes and lacks spines.
10. **Alders** are deciduous, and the leaves are alternate, simple, and serrated. The flowers are catkins with elongate male catkins on the same plant as shorter female catkins, often before leaves appear; they are mainly wind-pollinated, but also visited by bees to a small extent. These trees differ from the birches (*Betula*, another genus in the family) in that the female catkins are woody and do not disintegrate at maturity, opening to release the seeds in a similar manner to many conifer cones.
11. **Carob** grows up to 15 meters (50 feet) tall. The crown is broad and hemispherical, supported by a thick trunk with rough brown bark and sturdy branches. Its leaves are 10 to 20 centimeters (4 to 8 inches) long, alternate,

pinnate, and may or may not have a terminal leaflet. It is frost-tolerant to roughly -7°C (19°F). Most carob trees are dioecious and some are hermaphroditic, so strictly male trees do not produce fruit. When the trees blossom in autumn, the flowers are small and numerous, spirally arranged along the inflorescence axis in catkin-like racemes borne on spurs from old wood and even on the trunk (cauliflory); they are pollinated by both wind and insects. The male flowers smell like human semen, an odor that is caused in part by amines. The fruit is a legume (also known commonly, but less accurately, as a pod), that is elongated, compressed, straight, or curved, and thickened at the sutures. The pods take a full year to develop and ripen. When the sweet, ripe pods eventually fall to the ground, they are eaten by various mammals, such as swine, thereby dispersing the hard inner seed in the excrement.

Not included on this map but still on the land

12. **Ailanthus** or Tree-of-Heaven has rapid growth and can grow into a large tree, reaching heights of 80 feet and up to 6 feet in diameter. It is considered highly invasive and it's our most common tree (unfortunately). It is good for shade but not good to burn or smell.
13. **Chinaberry tree**, is a fast-growing deciduous tree that reaches 30 to 50 feet tall and has a canopy that is usually 20 feet in diameter. Also invasive, it is our second most common tree.
14. **Eucalyptus**.
15. Pepper Tree
16. **Evergreen** (christmas trees)
17. Redwood
18. Hachiya Persimmon
19. Fuyu Persimmon
20. Pomegranate
21. loquat
22. Peach
23. Guava
24. Orange
25. Lemon
26. Grapes

9.3. Final Soil Fertility Plan

How will you manage fertility in zones 2-5, as needed, throughout the seasons? (*Answer this question in the categories below and please specify plans for each zone*).

Site prep prior to planting:

(insert writing here)

Post-planting fertility management **Spring:**

(insert writing here)

Post-planting fertility management **Summer:**

(insert writing here)

Post-planting fertility management **Fall:**

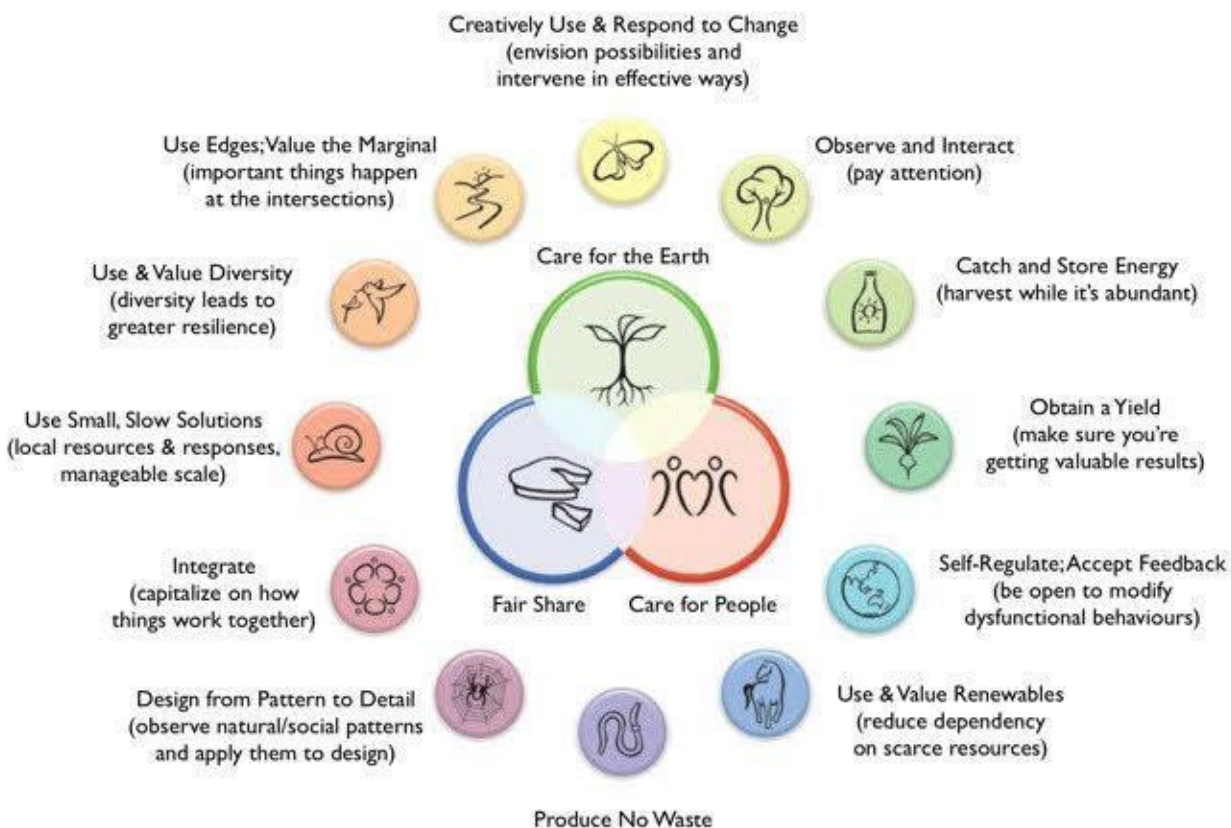
(insert writing here)

Post-planting fertility management **Winter:**

(insert writing here)

10. Final Design

10.1. Permaculture Principles



Observe and interact- Let's slow down and observe and appreciate our surroundings and others. Let's slow down and interact with others, in real life, not on our phones.

Catch and store energy- Take time to relax to restore balance and energy. Also, for me this means associating myself with those who bring positive energy and it means doing things that bring me joy and make me feel energized like running and learning.

Obtain a yield- In permaculture, this of course might mean planting in such a way that the most food is created. The food is the reward. In life, we can work to obtain an intrinsic or extrinsic reward. These rewards need not be strictly financial. Perhaps the reward is making a difference in your client's life or making your coworker's day a little bit easier.

Apply self-regulation and accept feedback- Self-regulation or self-control allows us to be accountable and it empowers us. Feedback helps us improve which will help us excel at work.

Use and value renewable resources and services and produce no waste- These two principles are simply common sense tips to living an eco-friendly lifestyle and in creating a more sustainable world.

Design from patterns to details- Permaculture calls for standing back and observing patterns in nature and in society and then filling in the details; this might make for a more efficient and logical way to work as well.

Integrate rather than segregate and use and value diversity- These principles translate perfectly in how we should all live in harmony with each other by integrating everyone into our communities and valuing diversity.

Use small and slow solutions- Permaculture uses small and slow solutions because they are easier to manage than bigger solutions and they allow for the use of local resources leading to more sustainable outcomes. While this may not always be feasible- sometimes we need big and fast solutions to solve some of society's greatest problems- this principle can potentially be a valuable tool for problem solving.

Use edges and value the marginal- Permaculture suggests that the interface between things is where the most interesting things take place. These are often the most valuable, diverse and productive parts of the system. This might mean extending a community garden out onto an edge near an unused railway. In life and work, we can try to more effectively use our resources including the edges.

Creatively use and respond to change- This requires no explanation, but in a changing world, it remains an important principle in both life and work. To remain a viable and sustainable business and to live well, we must be resilient and creatively work with change.

10.2. Final Design Phase Plan

To implement your design it will take time. Evaluate what will come first and why and how to continue the process to completion. Compile a list of actions and the order they will be completed in over the next 1-5 years.

Year 1: *(This year Fall/Winter) Cutting down Arundo. Planting bamboo. Getting rid of Atlantis trees. Digging out the pond again and obtaining pond liners. Finishing the community room so that the structure can hold solar panels. Build the first compost system with pallets. Build a rabbit enclosure. Build 2 chicken tractors.*

Year 2: *Cutting down some of the larger eucalyptus trees. Planting more sycamore and oak trees. Put solar panels on the community room (We are now a full solar farm). Build a cow enclosure. Redesign garden to star shape. Put up fences. Plant vegetable, herb, and flower gardens every year. Build 2 new humanure locations along with outdoor showers. Get rabbits. Put in greywater branch valve systems everywhere. Take out Atlantis trees. Add more beehives. Fill ponds with water. Plant nitrogen fixing trees. Build a second compost system in the garden. Expand Wwoofer program.*

Year 3: *Continue cutting eucalyptus and planting sycamore and oak. Get cows. Install rainwater harvesting systems everywhere. Take out Atlantis trees . Add more beehives. Build a third large compost system on the east side. Turn the garden into a Learning Garden that will invite the community to come and learn. Introduce Holacracy governance system.*

Year 4: *Clear trail for Lion Creek Path. Remove fallen trees in the creek, but still keep it wild. Build a flowform fountain. Re-landscape persimmon hill for more foot traffic. Build a gazebo in the center of the garden.*

Year 5: *(Social Permaculture elements). Create Timebank ™ for community participation. Create a Repair Cafe ™ . Host permaculture events. Create a Worker Space for the community. Create sister satellite Full Circle Farms ™ in other locations. Build a cob wall on the northern side to shield from neighbors.*

10.3. Social Permaculture Reflections

How have you taken local social issues into account in your design?

Full circle Farm has primarily focused on social issues throughout its history. One of the key tenants of our mission is social justice, democracy, shared responsibility and a voice for all. We have many sorts of meetings throughout the week and month. Every Thursday we have a general community meeting, where we sit in a circle, light a candle, hold hands and call in good intentions.

We also have conflict resolution meetings as needed which require mediation. We follow a nonviolent communication format, where we seek resolution.

We also have heart sharing circles. Here we sit in a circle and honor each other's gifts. This is not a time for discussing work or problems but purely just to honor each other's heart.

Full circle Farm started out patriarchal. Bob Goddard oh, the owner of the farm oh, pretty much had vetoing power oh, and the last word on anything, though it still felt like a democratic community. Bob was very fair and accommodating to everybody's needs. Which later evolved into a sociocracy model oh, and when I came aboard 6 years ago we were changing over to a holacracy model of governance. I personally like holacracy, and like the decentralized nature where everyone is a leader of their own domain and their own responsibility. But others found this very confusing and couldn't grasp the concept in its fullest. So there was always this call back to a patriarchal model. Right now we are struggling with a solid form of self-governance. Bob has retired from running the community oh, and since the beginning of covid spends most of his time at his son's residence up the road. Jaya is our current matriarch. She's also 83 years old, and has offered to step back from the more intense parts of running a community. so right now we're in another transition phase. I personally would like to see holacracy work to its fullest. The fact that Quail Springs is using it it's an inspiration.

11. Final Design Narrative/Map

You have now created individual final narrative/maps for each aspect of your final

design - it is time to now tell the full story. Use this section to either include a visual map or detailed written description of your final design with an emphasis on how all your elements and systems are in relationship to one another. What we are asking you for here is a whole system analysis. After working on the individual layers of your final design, now you want to demonstrate the web of relationships that you built in this final design guided by the permaculture ethics and principles.

Remember:

Describe the permaculture process. What are the steps you take to create a permaculture design?

The permaculture process involves observation, information gathering & research, creating a conceptual design (element, zones, sector analysis etc.), creating a final design, implementation and evaluation.

Describe your project vision, restating your mission and goals for the project.

The vision of Full Circle Farm is to live in harmony close to the land in health and caring for each other. we believe in permaculture organic farming, self-governance, sovereignty, and happiness. to this end we adopt permaculture principles for the land.

The 5 acre property where 20 of us call home originally started out as a place where people could help rebuild the burnt down house, and in exchange live here for free. That concept has evolved over the last 35 years to the present Full Circle Farm community. At first it was pretty random. Everything was just placed haphazardly on the farm. Shacks were built without regard for building codes, or the direction of the Sun or convenience. people just lived in random places and trailers and motorhomes.

Over the last 30 years we have found where things grow best, where the sunlight shines best, and where the wind and water flows. This permaculture design project takes all of this knowledge and experience into account. One of our big goals is to remove a lot of the invasive trees that were planted years ago, and replace them with drought tolerant native plants. Another goal is to bring back the farm animals that we've had here, and incorporate them into that permaculture food cycle.

Finally, we want to create a barrier around the perimeter of the farm so that our neighbors don't look in. This involves different edges and tree barriers and cob walls.

We want to revive our holacracy governance system and make it work better than it did before. Eventually we want to create another Full Circle Farm, on another piece of land.

It will have the same principles and guidelines as this one. From there we can branch out and create even more Full Circle Farms! The goal is to spread a good idea to everyone!

Write a paragraph to explain how you have applied the principles of stacking functions, planned redundancy, and relative location to your design.

We have multiple sources of water: city water, rain water, well water. We now have multiple uses for water discharged from sinks showers purifiers. We have bees that provide honey and pollinate flowers. We have a good compost system that provides food for chickens and soil for our garden. We recycle metal and glass. We repair our vehicles here on site. We reuse. We pickle extra vegetables and make a kombucha. Also we have grains and legumes that will last us four months stored in our basement. If the world were to end tomorrow we would function pretty well for months!

Explain the permaculture ethics and how they have guided your design.

*The permaculture ethics of **Earth Care, People Care and Fair Share**. They have guided our design in many ways.*

Earth Care: *Because we don't know the levels of our well water, and fear that eventually it might go dry, Full Circle Farm uses a water conscious design that is climate appropriate and regionally aware. We will begin to use low water farm techniques, such as Grow Biointensive™. We will take out all the Eucalyptus trees and put in sycamore. Integrating perennials into the design so as to be water wise and to allow succession Increased rainwater harvest and gray water basins- the hydrophobic soils are prone to alot of runoff before water has a chance to infiltrate. Our implementation of various swales, sunken beds and channeling of greywater bring and keep water in the most vegetated areas of the site. Earthworks and water harvesting is an Earth Care practice.*

People Care *is about governance and social justice. It is reflected in our community. We looked out for each other as a family would. If somebody is sick we take care of them, if somebody needs a job everybody helps to get that person employed. We also Help each other with plant medicines if someone is sick.*

Fare Share *Is reflected at Full Circle Farm in that we share many things. We have a workshop or we share tools. We share our gardening tools also. We even share Vehicles sometimes. Everybody does their fair share of work around the farm too. Finally we share food, and we share cooking of the food every night. What we don't use on the farm goes*

to the local homeless shelter or food bank.

Part 4

12. Design Review & Commentary (required for certificate)

Your workbook so far has been like a work-in-progress journal for your personal project as you work your way through the PDC curriculum. If you wish to get your final design reviewed & commented on to receive a certificate, please insert/paste your final design maps and layers below.

Note:

- Our reviewers (as well as potential clients, contractors, landscape architects, etc.) should ideally be able to clearly understand the detailed design and all its interrelated systems and implementations without much text or explanations.
- The design should ideally be able to stand on its own, especially with the individual layers accompanying it.
- Thus, please double-check essential descriptive details (e.g. scale, orientation, key, etc.) on all maps, layers, and plans; and make sure all maps and layers 'fit' and work together to form a coherent and sensible whole.
- If something does not seem to make sense or there is a 'type 1' design error, a reviewer will try to clarify any potential misunderstandings, and *might* go back into your workbook to find an explanation or reasoning behind a design decision.
- Please keep in mind, with the amount of designs to review, it is not possible for our reviewers to go through every workbook in detail and can primarily focus on the final part 4.

For a design review we require:

12.1. [Original Basemap](#)



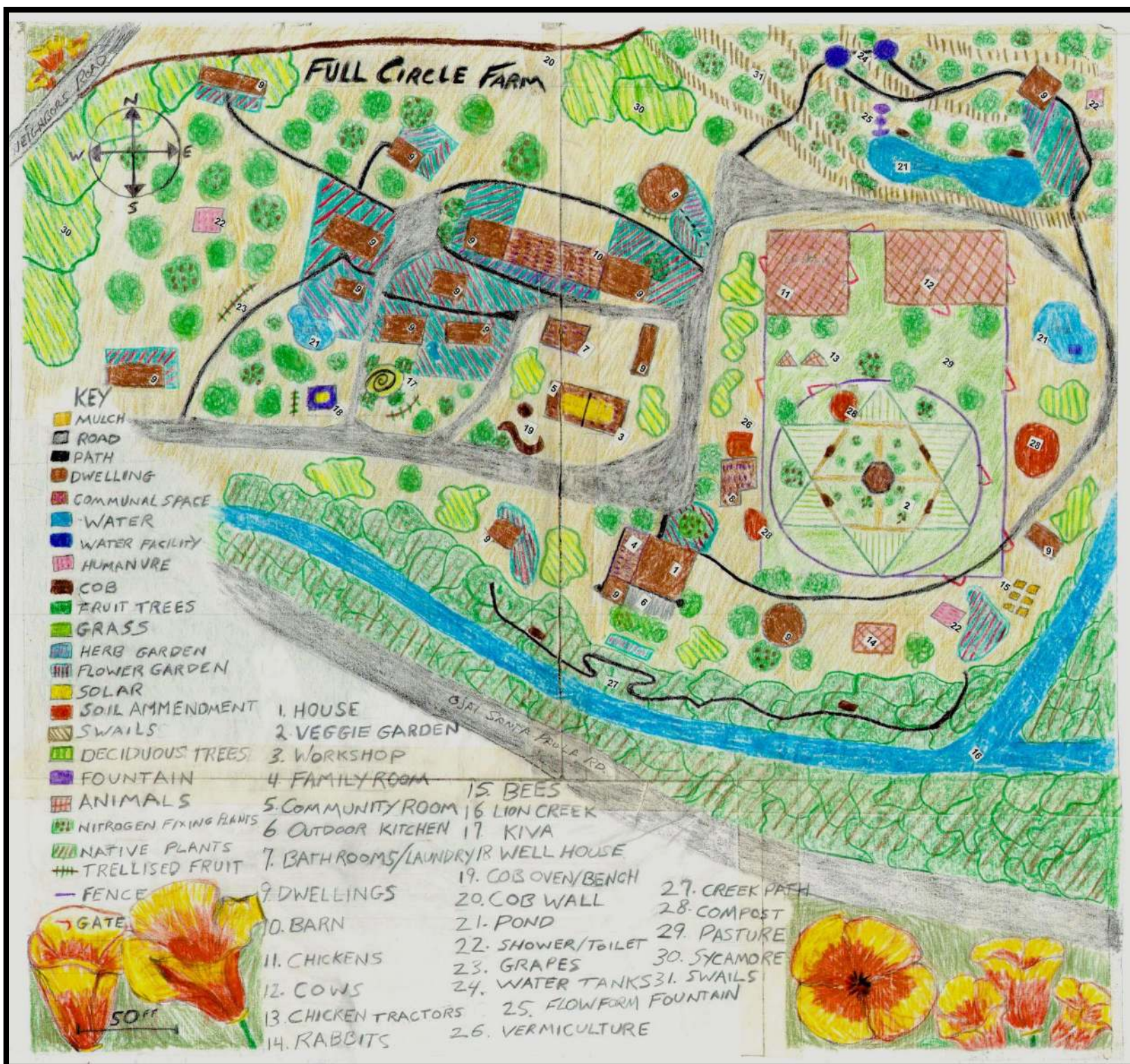
12.3. [Sector Analysis layer/map](#)



12.4. [Water/slope/elevations layer/map](#)



12.5. Final detailed design layer/map



12.6. Final Zone layer/map



(Here are links to all the other maps throughout this document!)

[Contour Map](#)

[Current Plant Map](#)

[Zone 1 Design Map](#)

[Final Water Map](#)

[Microclimate Map](#)

[New Zones and Pathways Map](#)

[Final Vegetation Map](#)



This design was a collaboration between Robert Redecker and Andrew Ranjbaran of Full Circle Farm.

Congratulations! Your individual design project is now complete. Email your completed workbook to Aris (pdc@quailsprings.org) by 10pm (PST) on October 17th, 2022. The Quail Springs design project review team will be reviewing your project shortly, giving you feedback, and sending out information about certificates.